

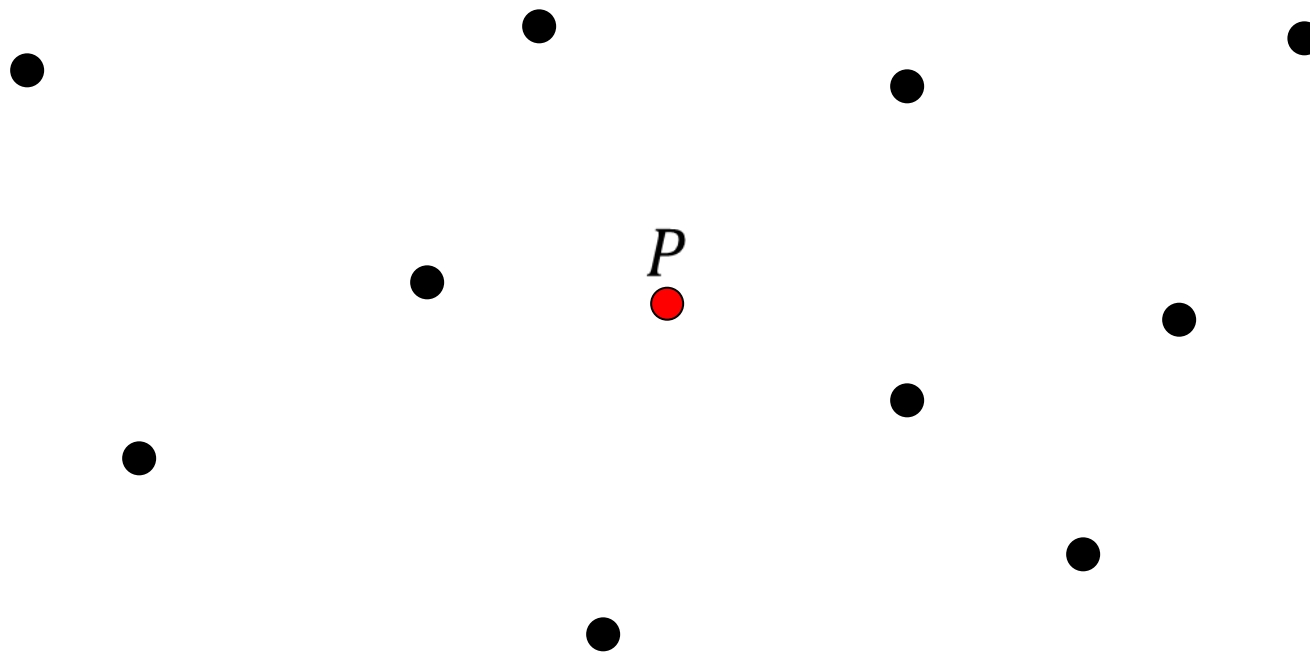


Voronoi Diagrams and Delaunay Triangulation

One Can Be Created From the Other

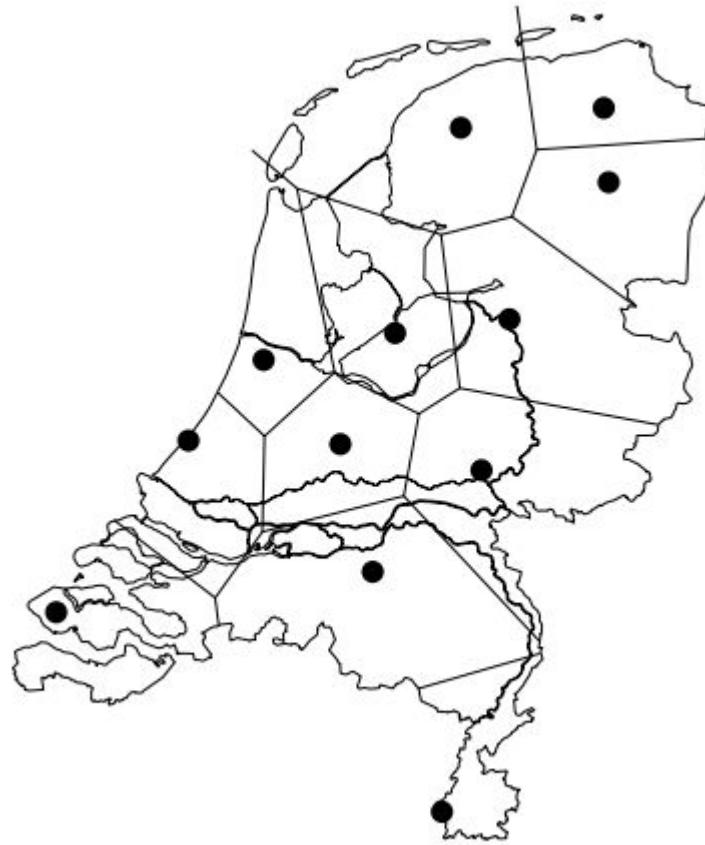
Post Office Problem

- ▶ Given a set of post-office locations and an arbitrary location P , find the post-office that is closest to P



Post Office Problem

- Given the locations of post offices in a city, how do we optimally assign a post office for each residence where they will receive their mail?



Voronoi Diagram

- ▶ A Voronoi Diagram is a geometric structure that consists of a set of points called *sites*

A ●

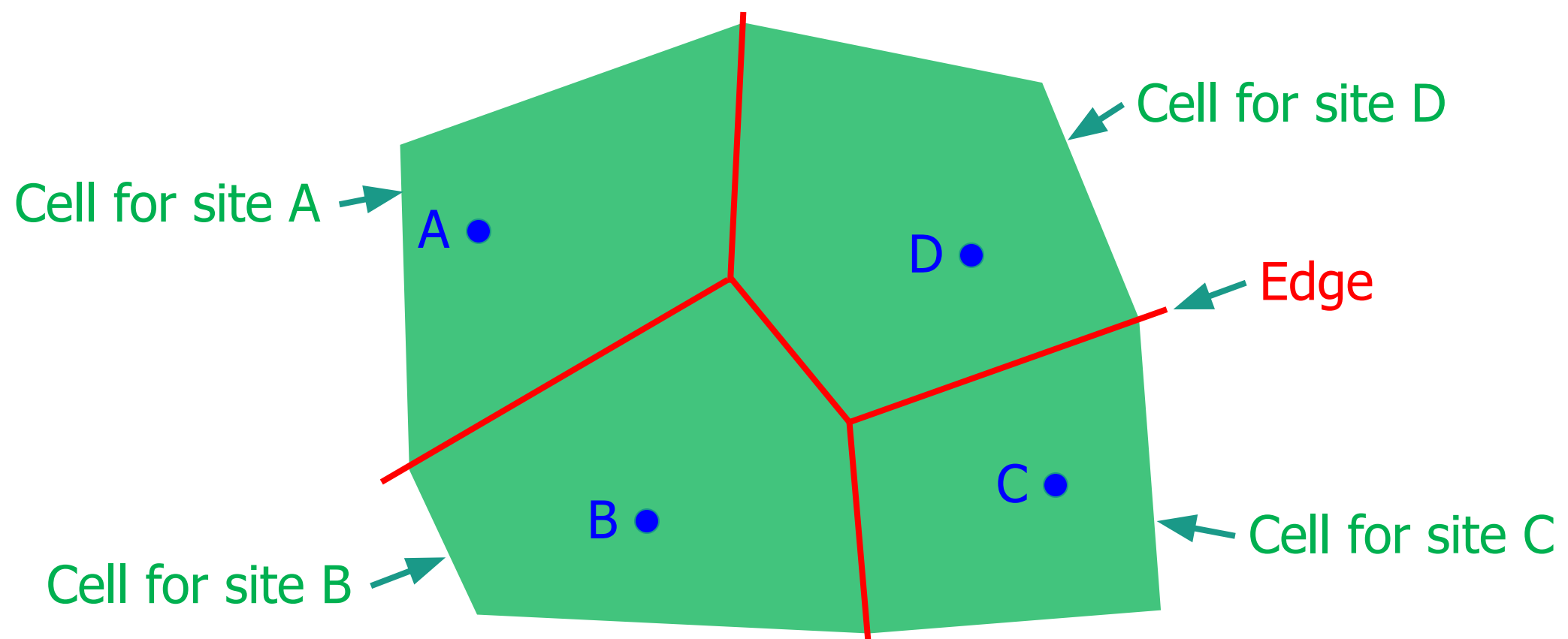
D ●

B ●

C ●

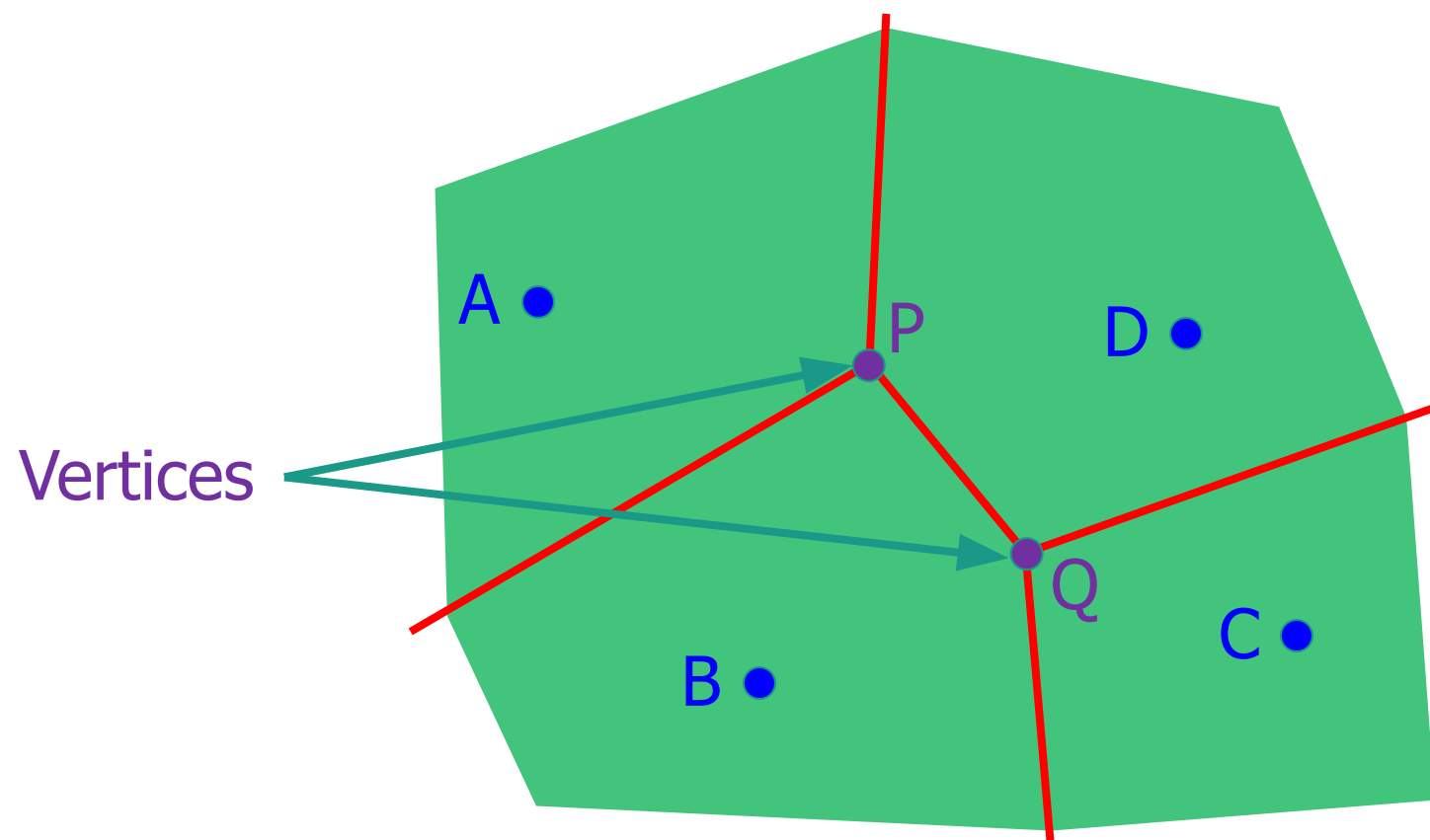
Voronoi Diagram

- ▶ A Voronoi Diagram is a geometric structure that consists of a set of points called *sites* that are assigned to regions or *cells* bounded by *edges*



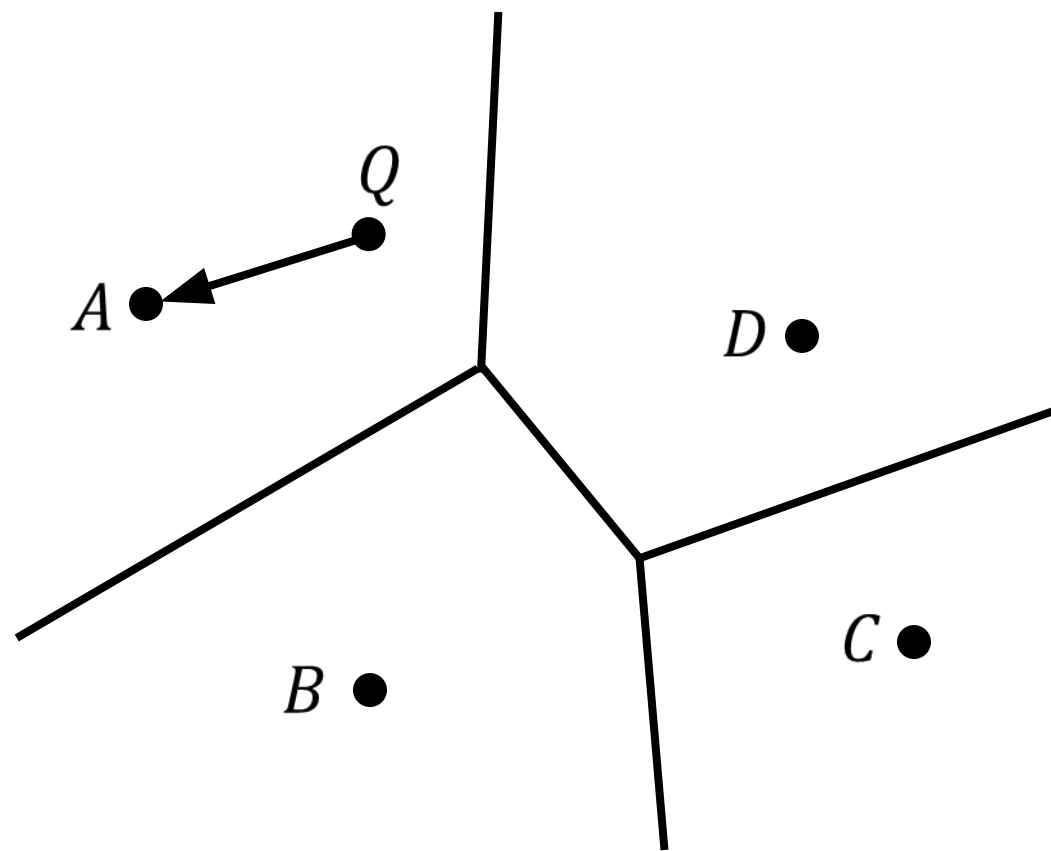
Voronoi Diagram

- ▶ A Voronoi Diagram is a geometric structure that consists of a set of points called *sites* that are assigned to regions or *cells* bounded by *edges* that intersect at *vertices*



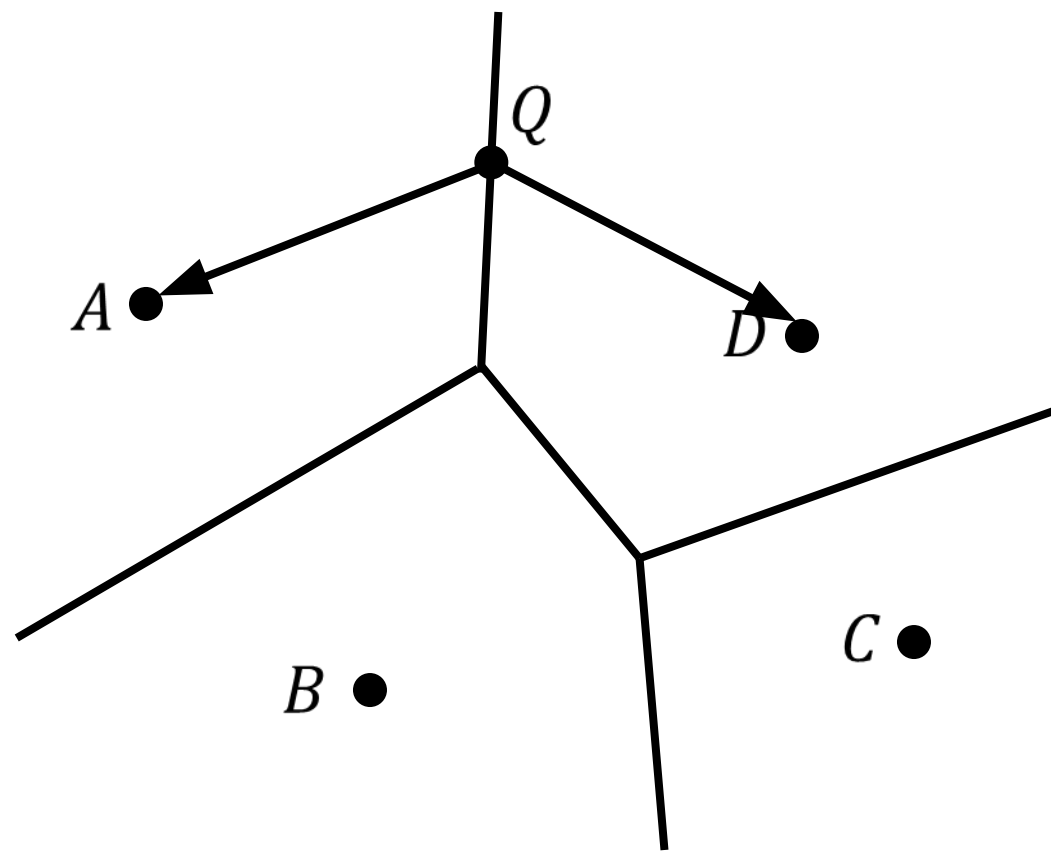
Voronoi Diagram

- ▶ Voronoi Diagrams are structured such that any point Q within a cell is closest to the site that is assigned to that cell



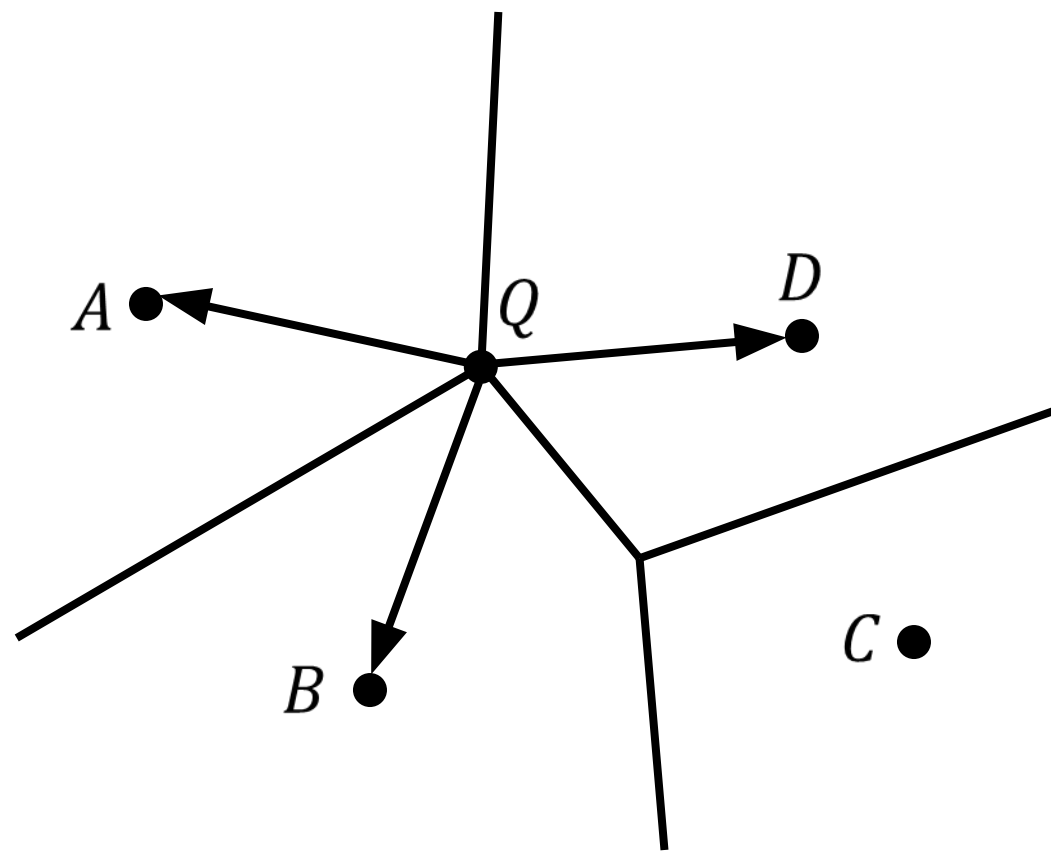
Voronoi Diagram

- If a point Q is along an edge that separates two sites, then Q is equidistant from the two sites



Voronoi Diagram

- If a point Q is on a vertex, then Q is equidistant to the sites that are adjacent to that vertex



Voronoi Diagram

- ▶ Given a set of distinct points/sites P on a plane, the Voronoi diagram of P , denoted by $Vor(P)$, is a subdivision of the plane into n cells, one for each site in P , where an arbitrary point Q lies on the cell corresponding to site P_i if and only if

$$\text{dist}(Q, P_i) < \text{dist}(Q, P_j) \text{ for each site } P_j \in P, j \neq i$$

- ▶ $\text{dist}(A, B)$ is the function for measuring the distance between two sites A and B

Voronoi Diagram

- ▶ The most common measure for distance is the Euclidean distance, which is defined as

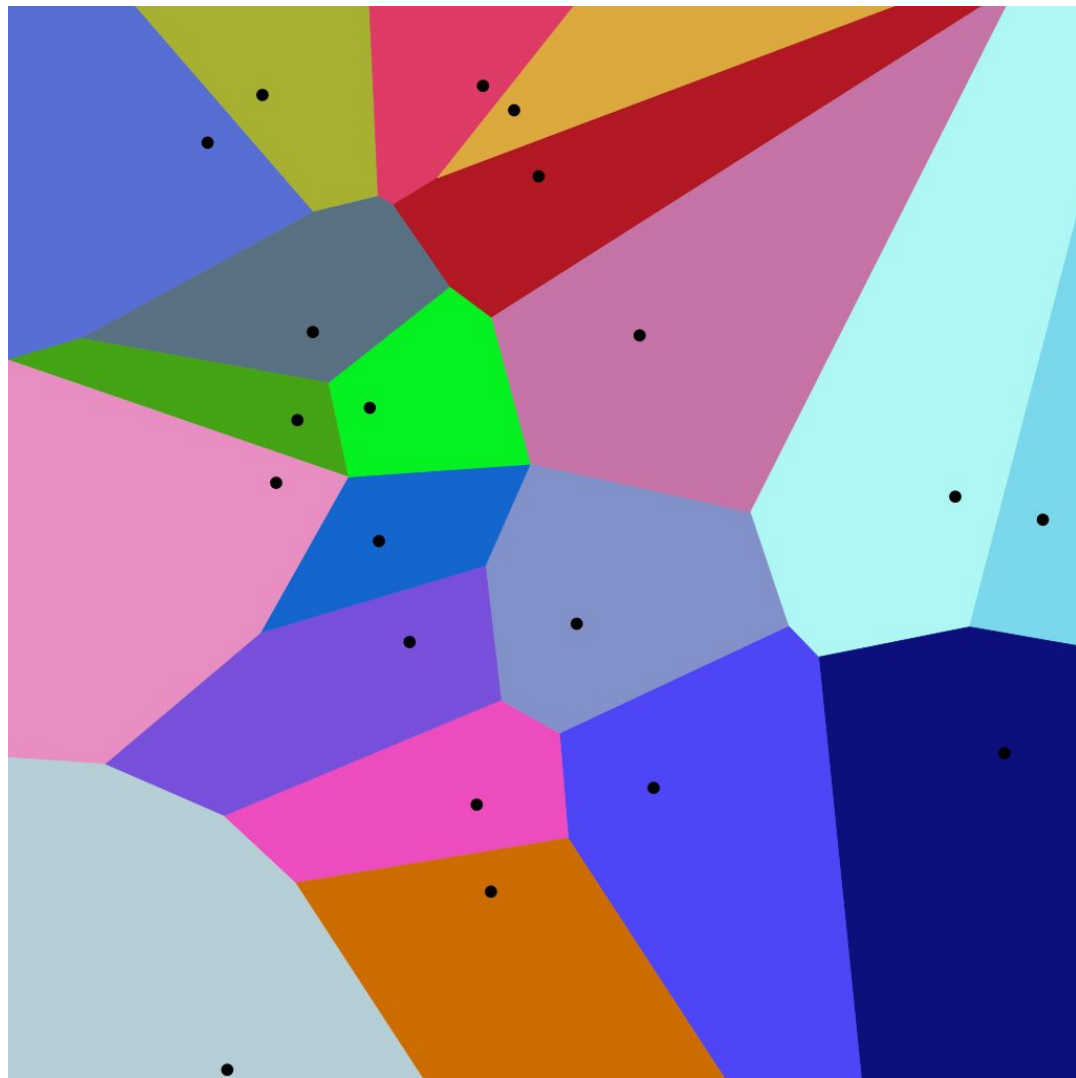
$$\text{dist}(A, B) = \sqrt{(B_x - A_x)^2 + (B_y - A_y)^2}$$

- ▶ Another measure for distance is the Manhattan distance, and is defined as

$$\text{dist}(A, B) = |B_x - A_x| + |B_y - A_y|$$

- ▶ For our class, we will assume Euclidean distance

Voronoi Diagram



<https://commons.wikimedia.org/w/index.php?curid=38534275>

Voronoi diagram using
Euclidean distance



<https://commons.wikimedia.org/w/index.php?curid=38535751>

Voronoi diagram using
Manhattan distance

Voronoi Diagram

- ▶ Voronoi Diagrams present a way to encode proximity information, which can be used to quickly perform queries such as
 - ▶ Which object is closest to some point P
 - ▶ Where is the nearest hospital/grocery/mall from an arbitrary location P in an area?
 - ▶ Which cell phone tower is nearest to a phone that is located at some location P ?
 - ▶ What restaurants are near my current location?

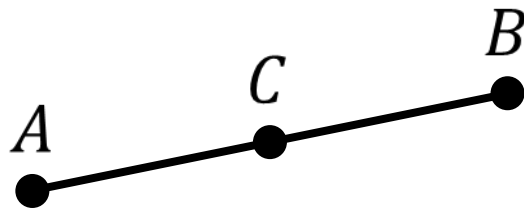
Voronoi Diagram

- ▶ Going back to the post-office problem, we can create a Voronoi diagram, and assign residences to post-offices based on the cells (Voronoi cells)
- ▶ Voronoi assignment model



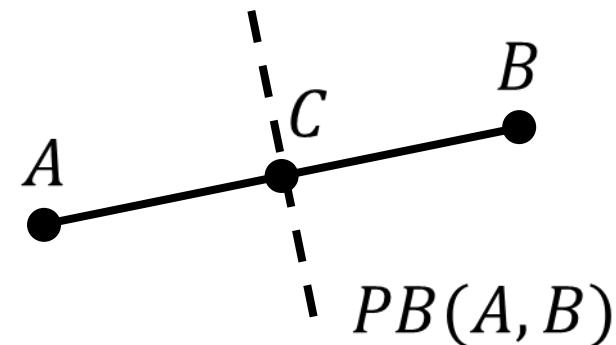
Constructing Voronoi Diagrams

- ▶ Voronoi edges are constructed from *perpendicular bisectors* of the lines that connects two sites
- ▶ Given two sites A and B , the perpendicular bisector $PB(A, B)$ can be obtained as follows:



Find midpoint C and slope m_{AB} of the line from A to B

$$C = \frac{A + B}{2}, m_{AB} = \frac{B_y - A_y}{B_x - A_x}$$

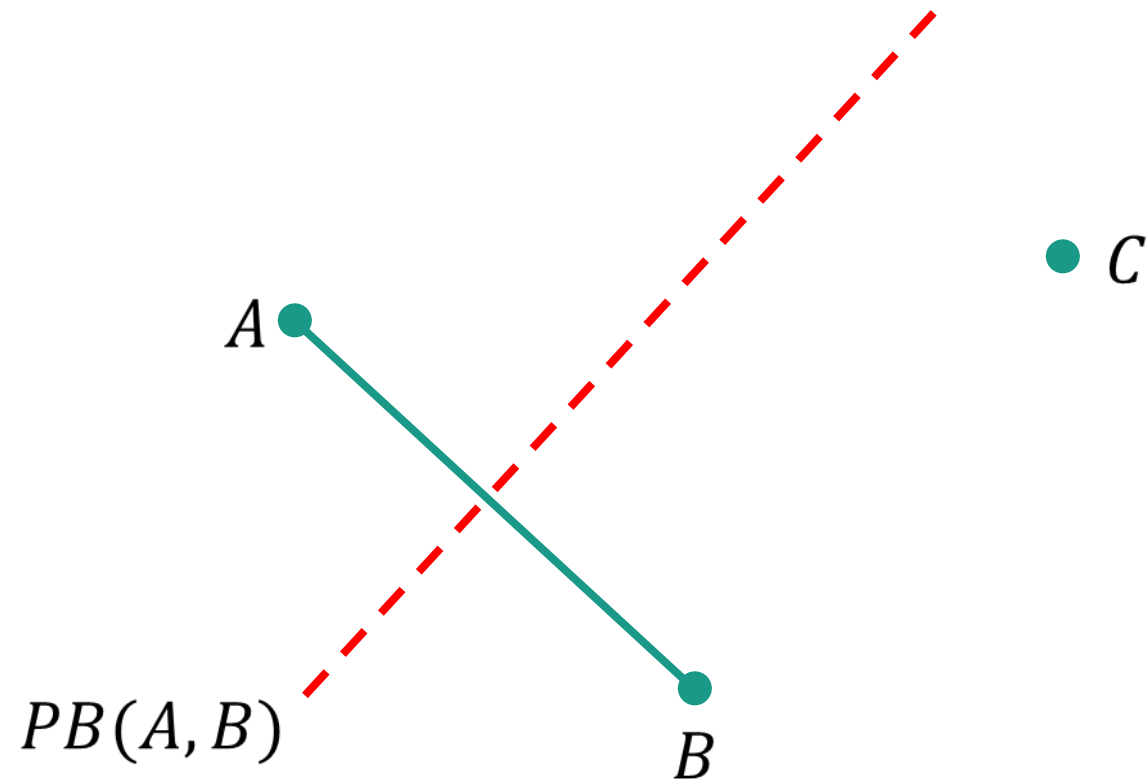


Generate the equation of the line for the perpendicular bisector

$$(y - C_y) = -\frac{1}{m_{AB}}(x - C_x)$$

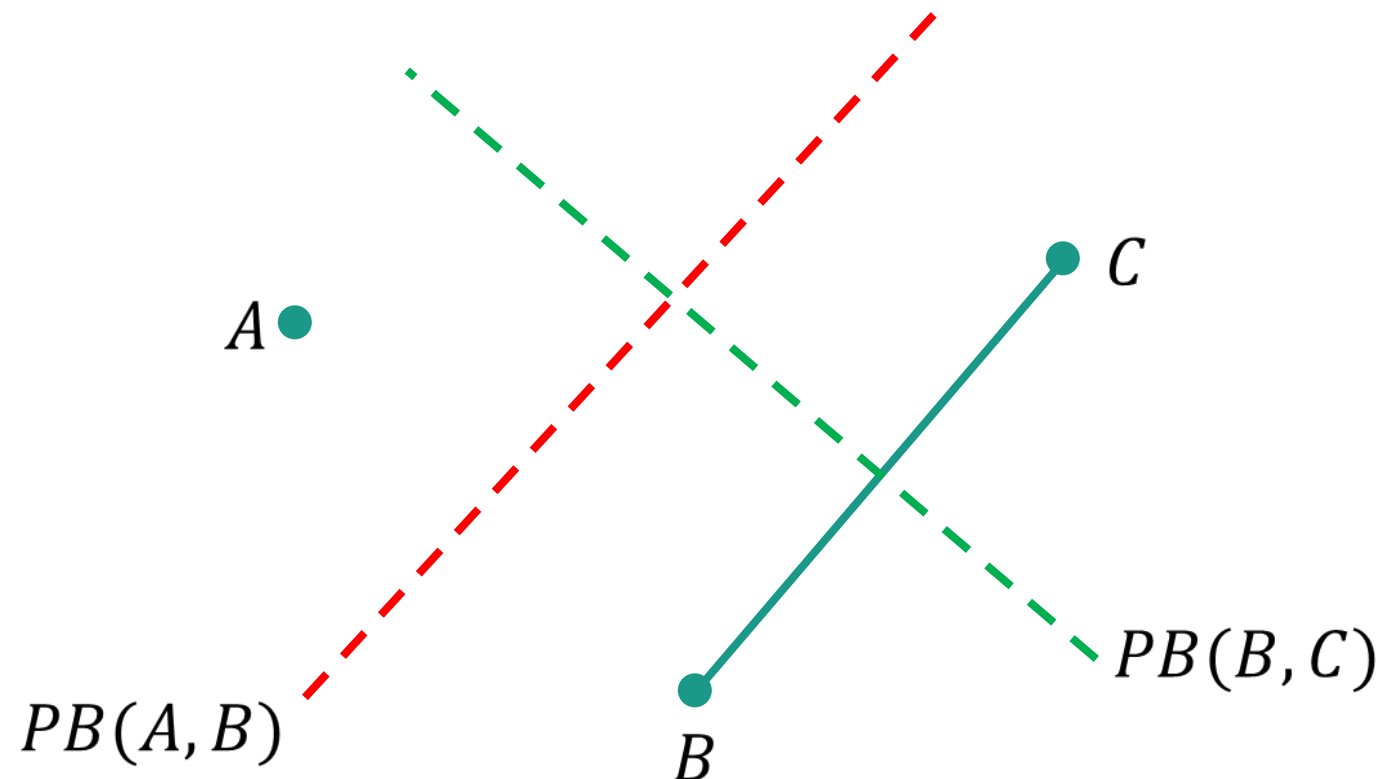
Constructing Voronoi Diagrams

- ▶ What about with 3 or more sites?
- ▶ First, construct the perpendicular bisectors for each pair of points



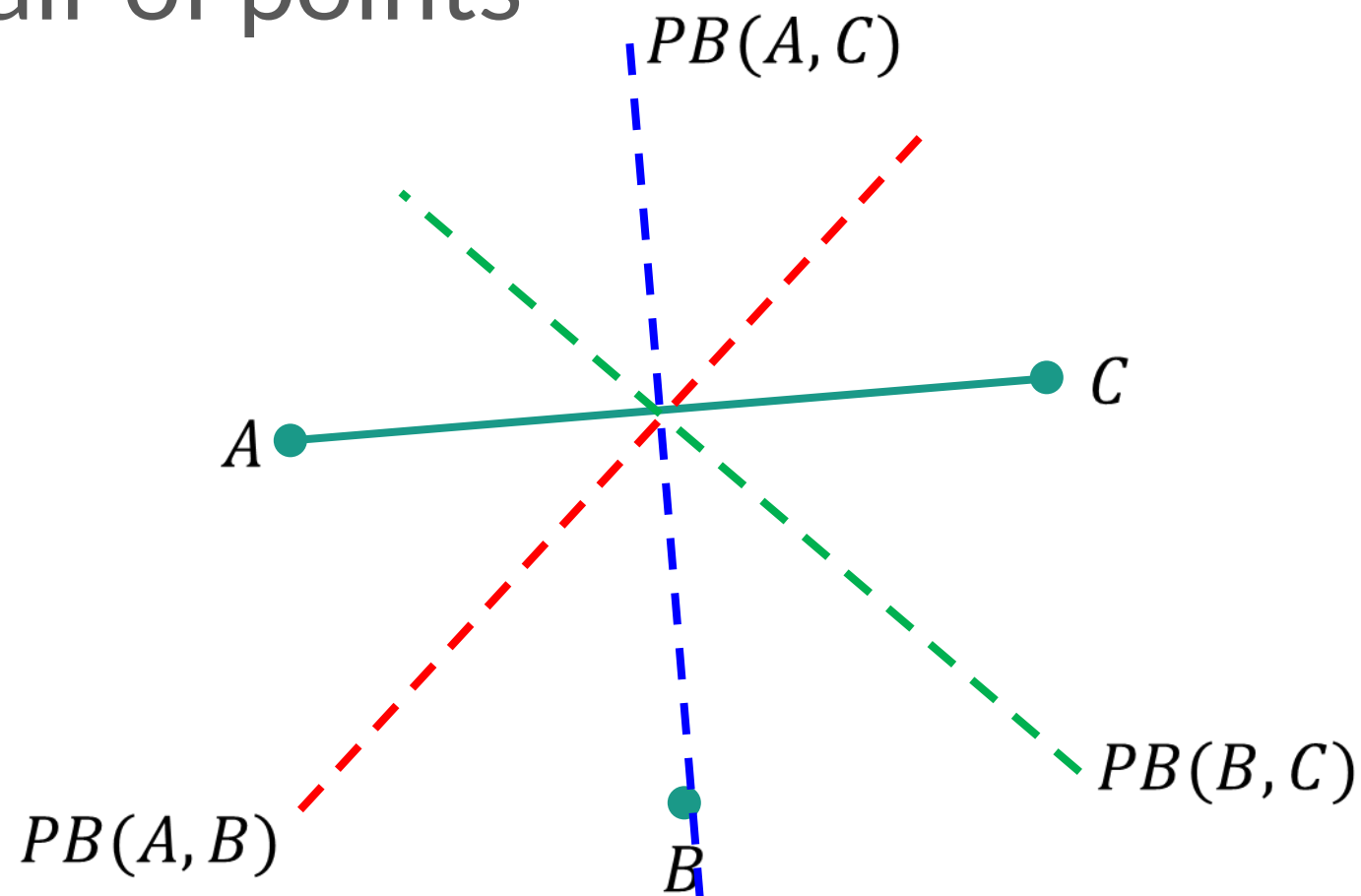
Constructing Voronoi Diagrams

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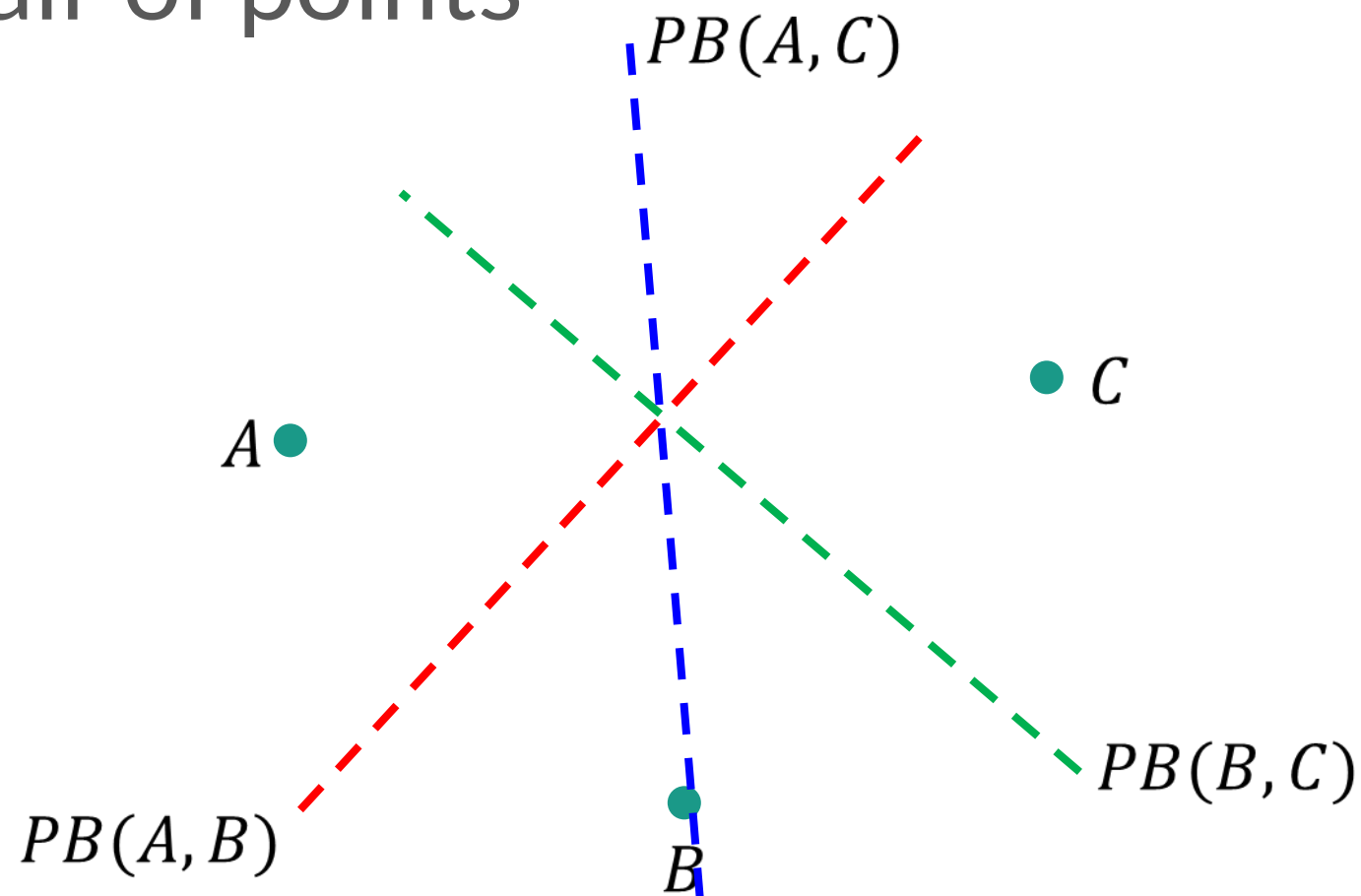
Constructing Voronoi Diagrams

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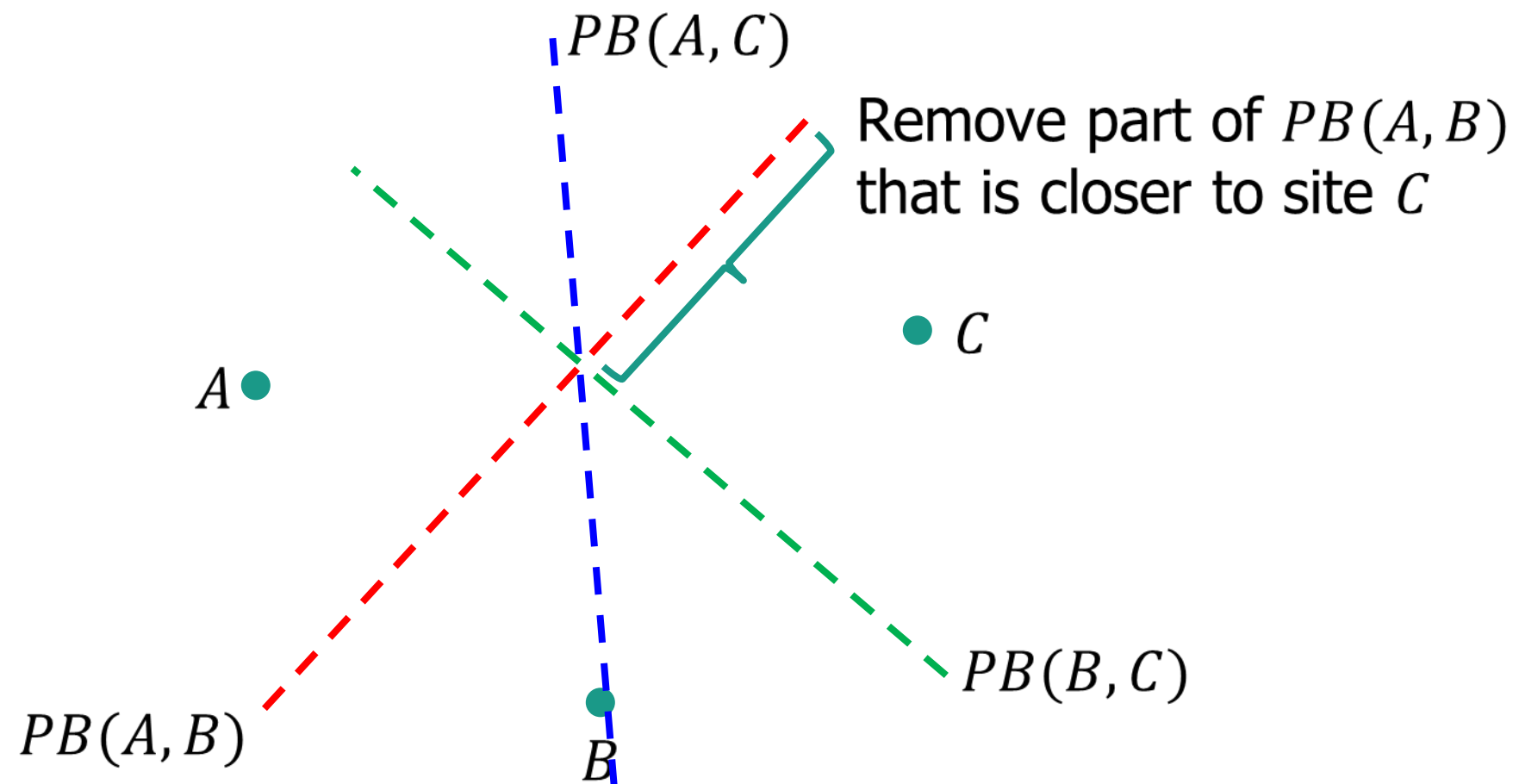
Constructing Voronoi Diagrams

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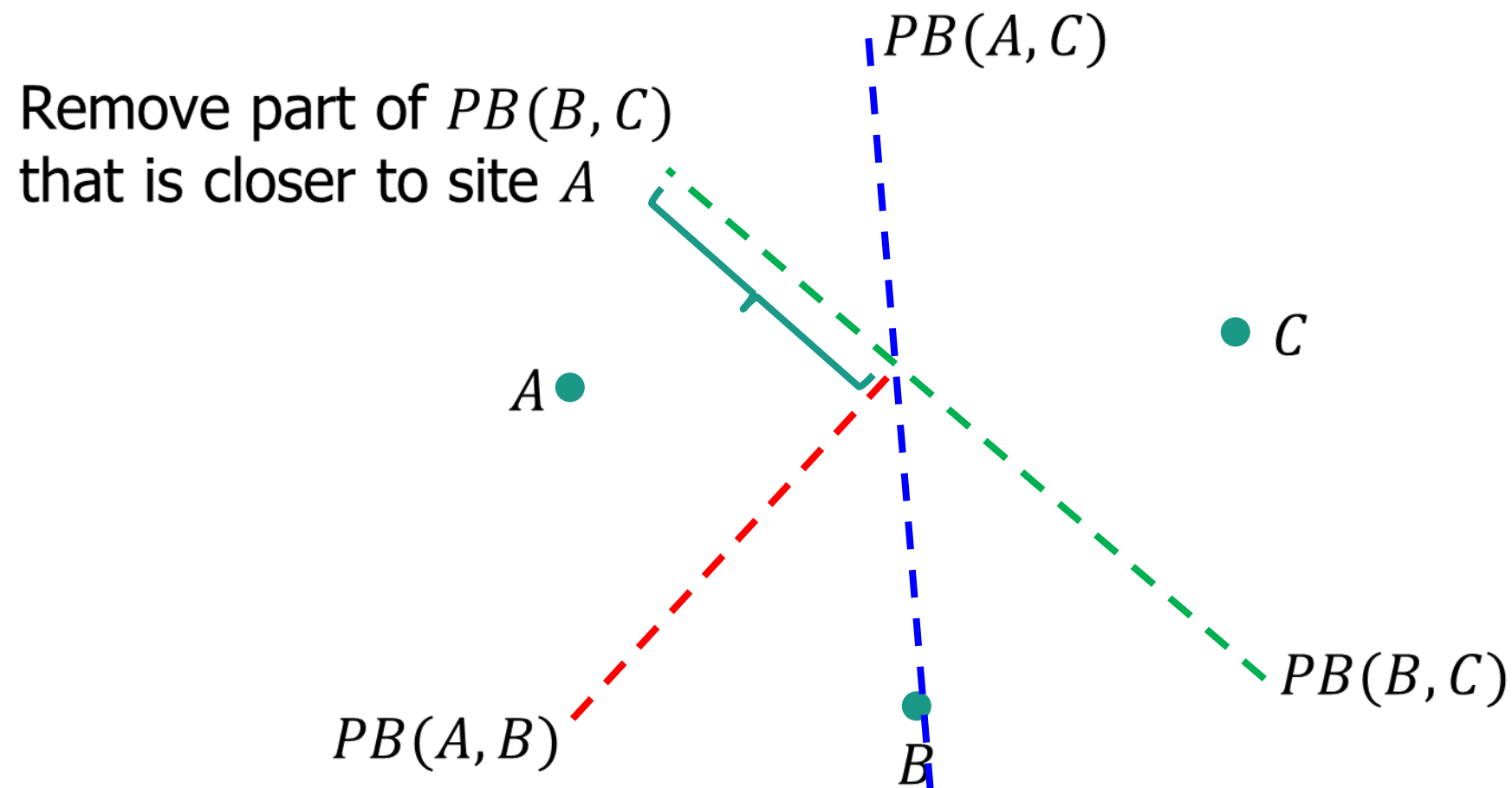
Constructing Voronoi Diagrams

- For each perpendicular bisector, remove the part of the line that is closer to a third site



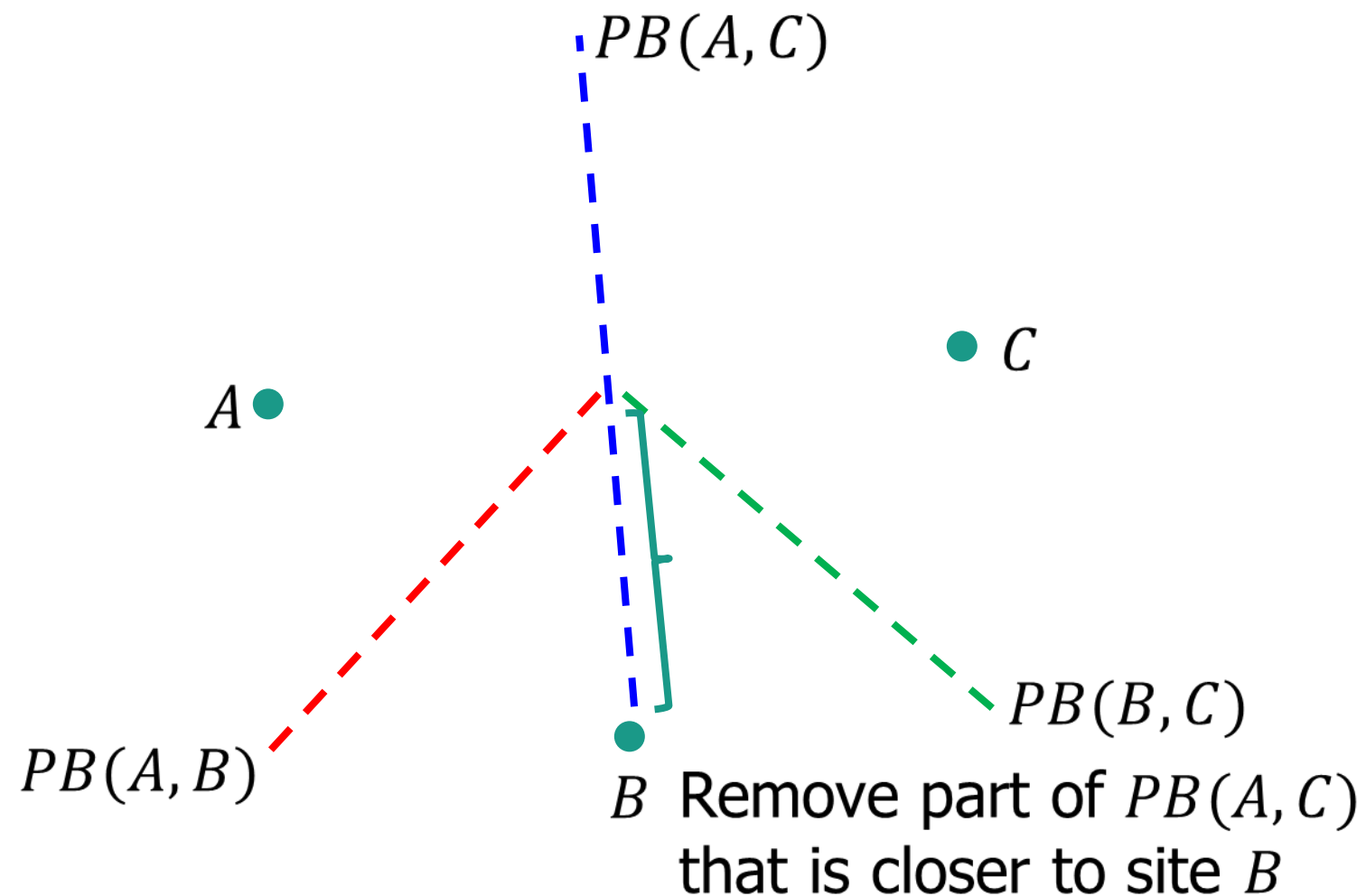
Constructing Voronoi Diagrams

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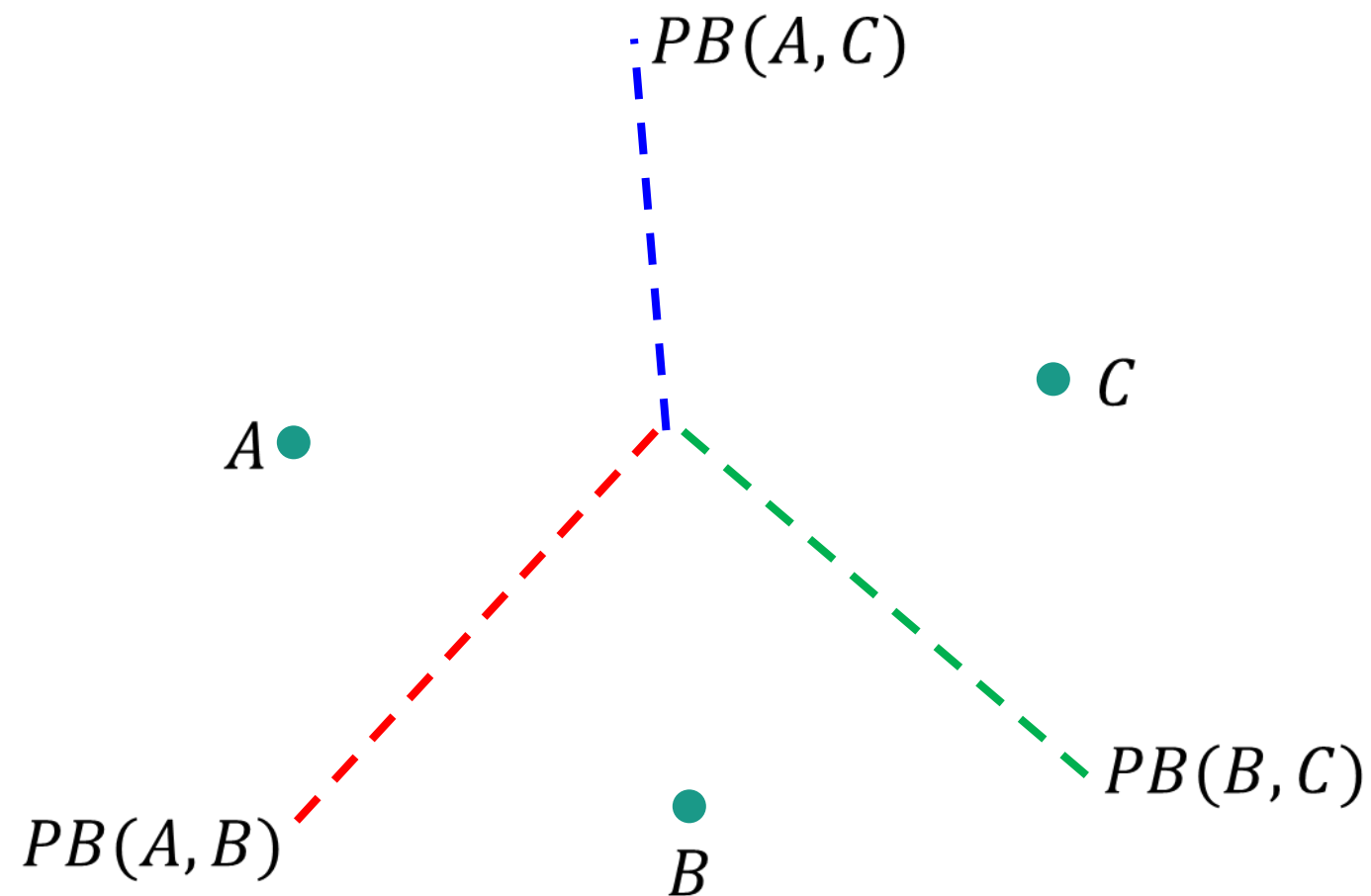
Constructing Voronoi Diagrams

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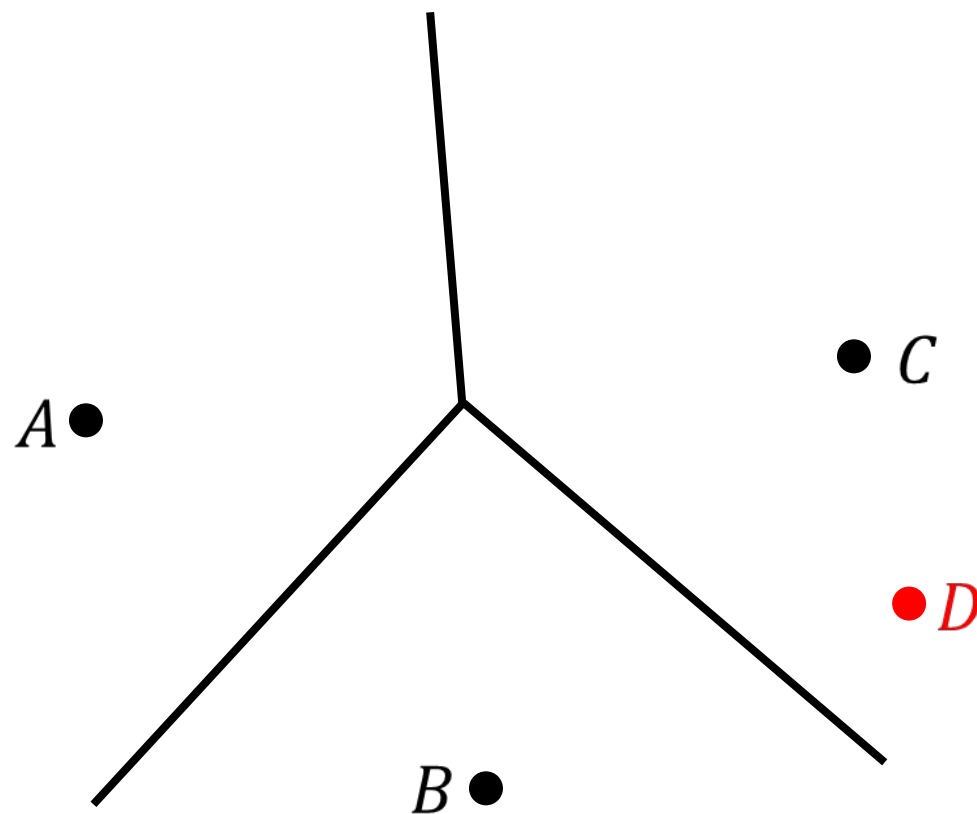
Constructing Voronoi Diagrams

► Resulting Voronoi diagram



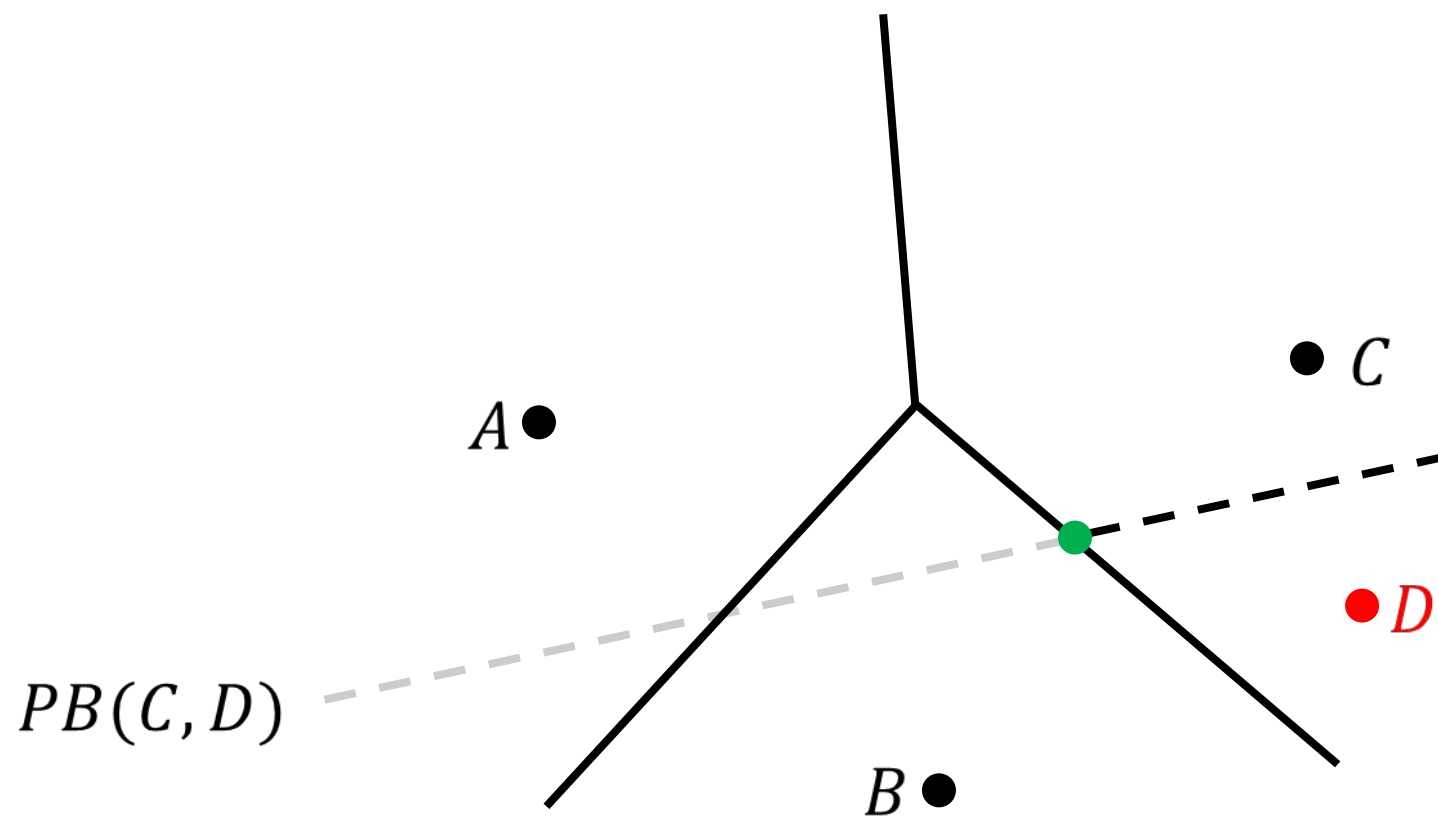
Adding a New Site

- Let's say we want to add a new site D from an existing Voronoi diagram with sites A , B , and C . How do we update the Voronoi diagram to include the new site D ?



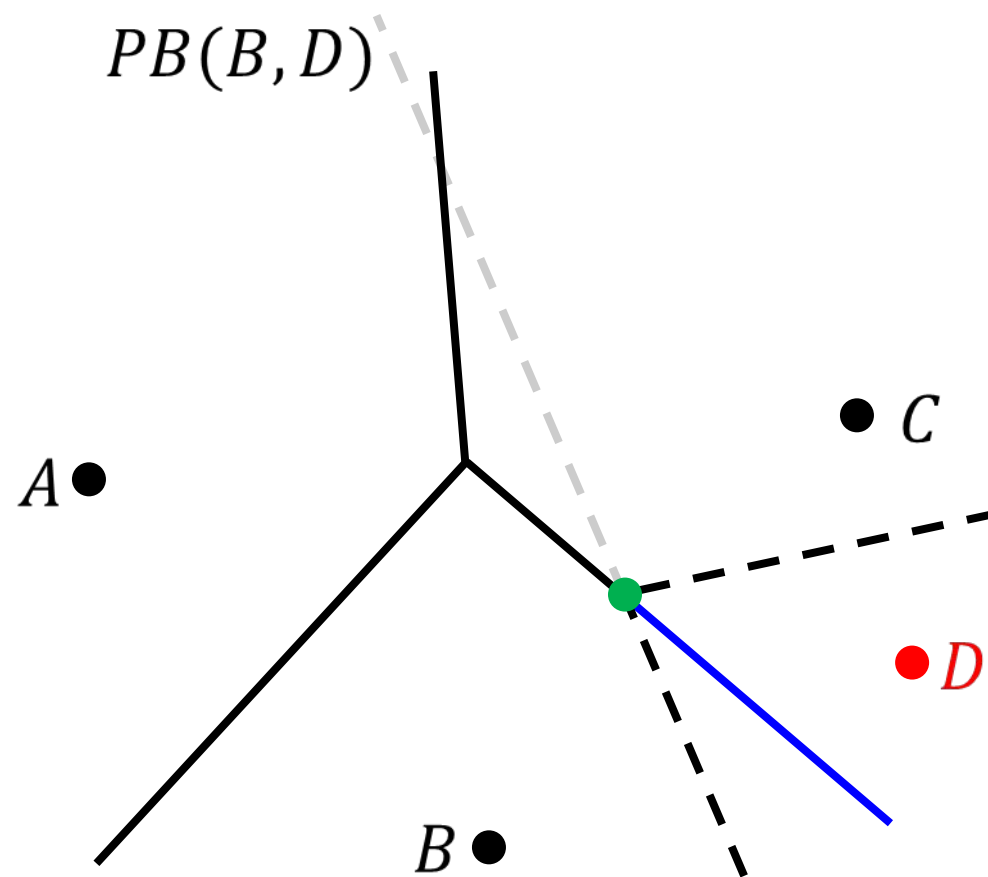
Adding a New Site

- Determine the cell that contains D (site C in this case). Construct the segment of the perpendicular bisector $PB(C, D)$ that is inside cell C. Segment intersects with an edge, so create a vertex at the intersection point



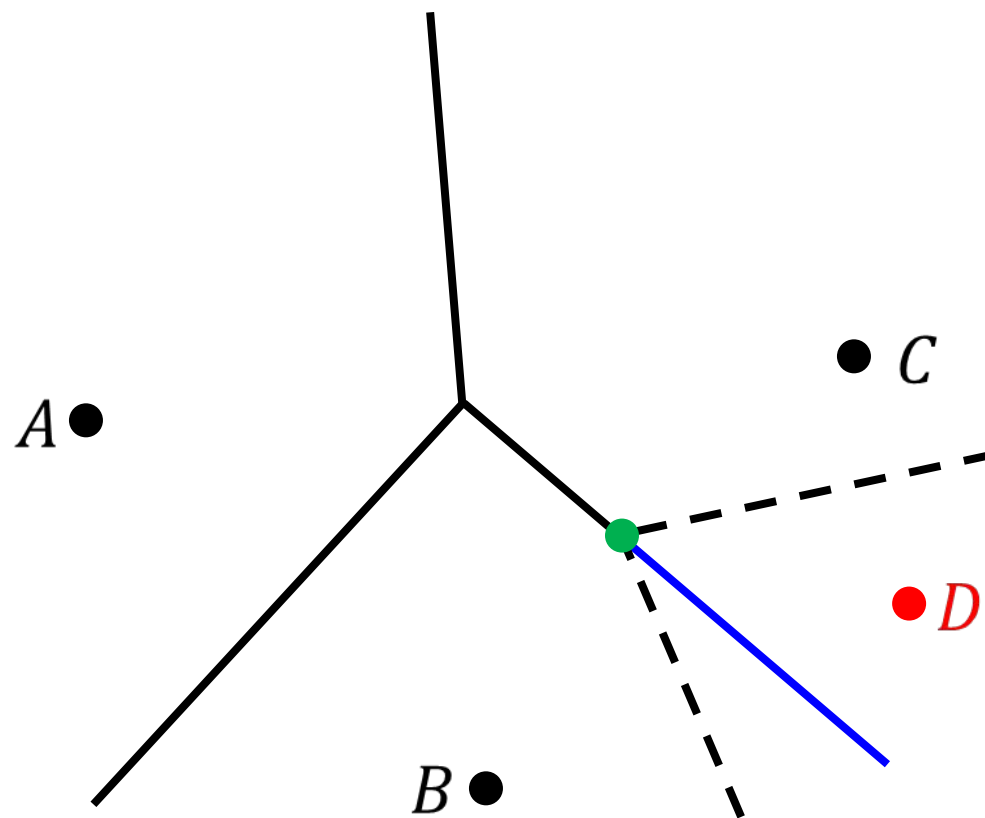
Adding a New Site

- ▶ Since the vertex is adjacent to site B , construct $PB(B, D)$.



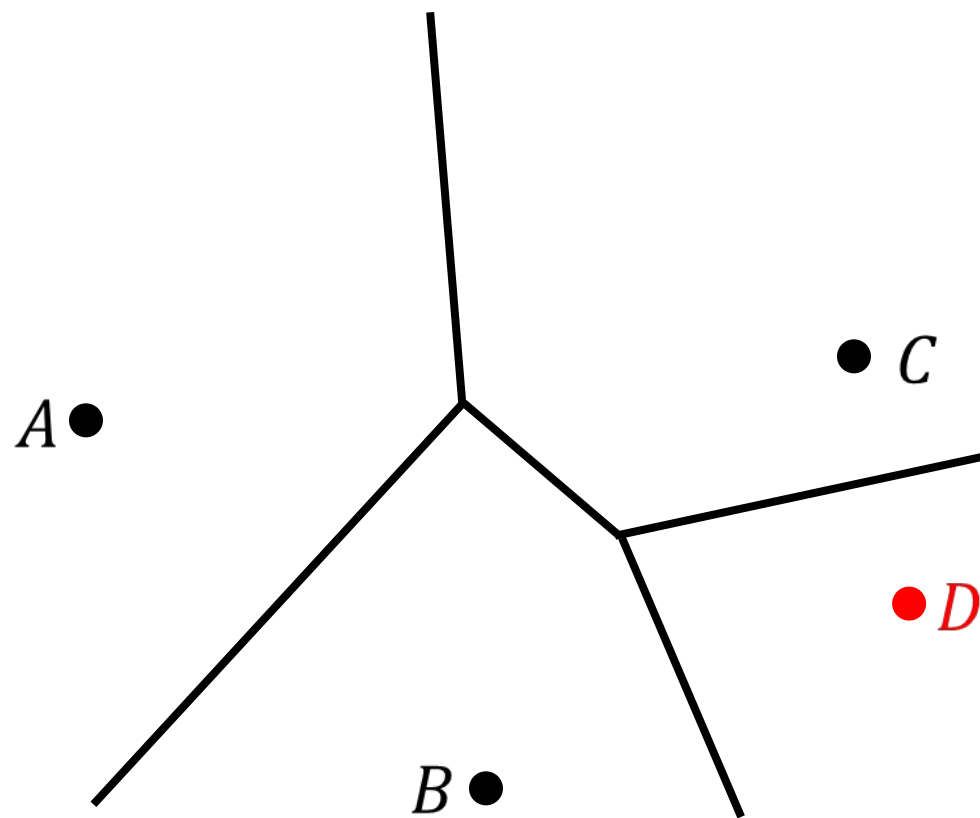
Adding a New Site

- Since no new vertices have been created, we remove segments of existing edges that are now inside the cell assigned to D



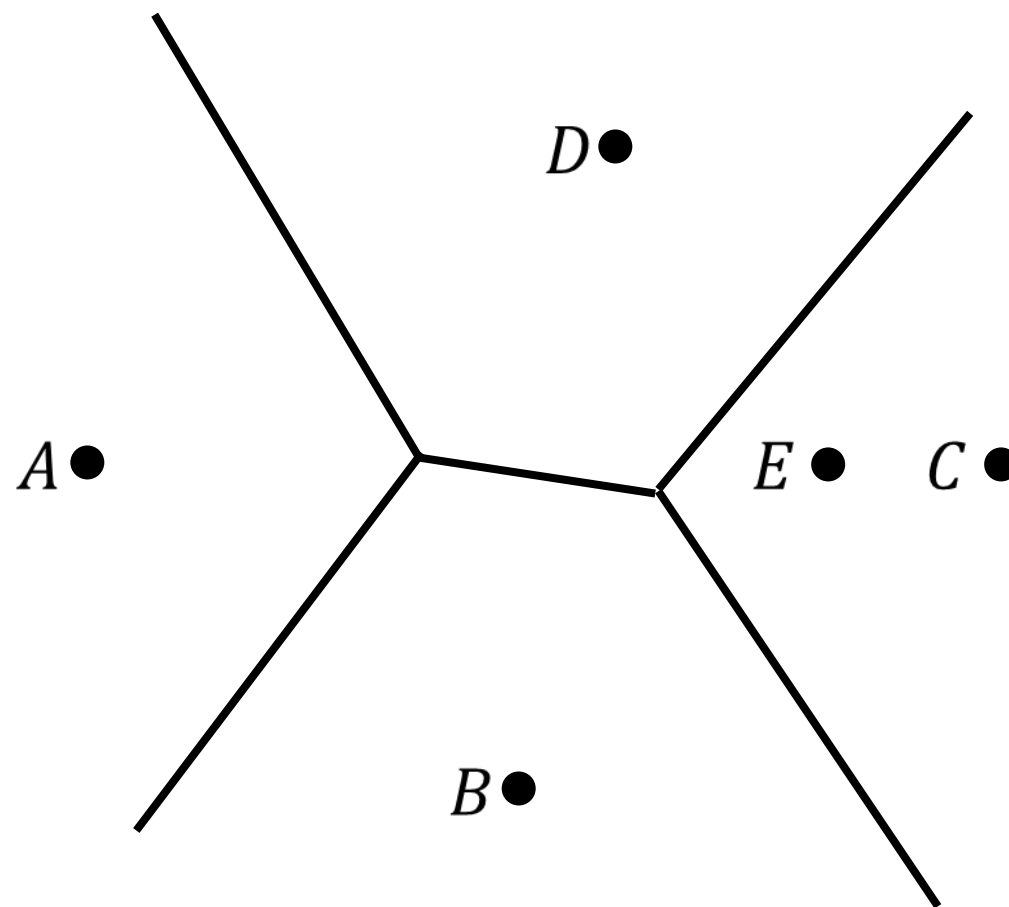
Adding a New Site

- Resulting Voronoi diagram after adding the new site D



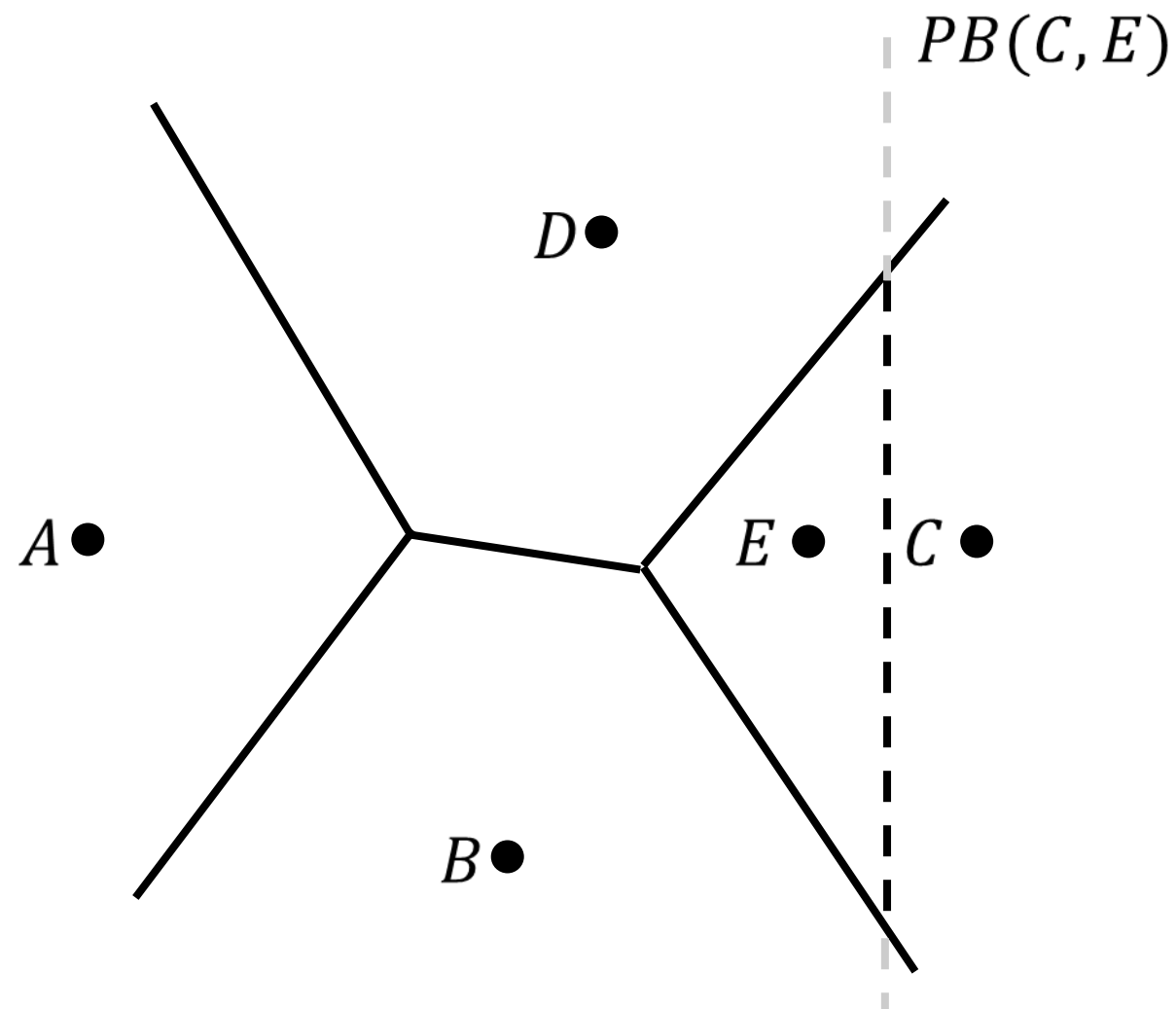
Adding a New Site

- ▶ Another example: Given a Voronoi diagram with sites A , B , C , and D , we want to add a new site E



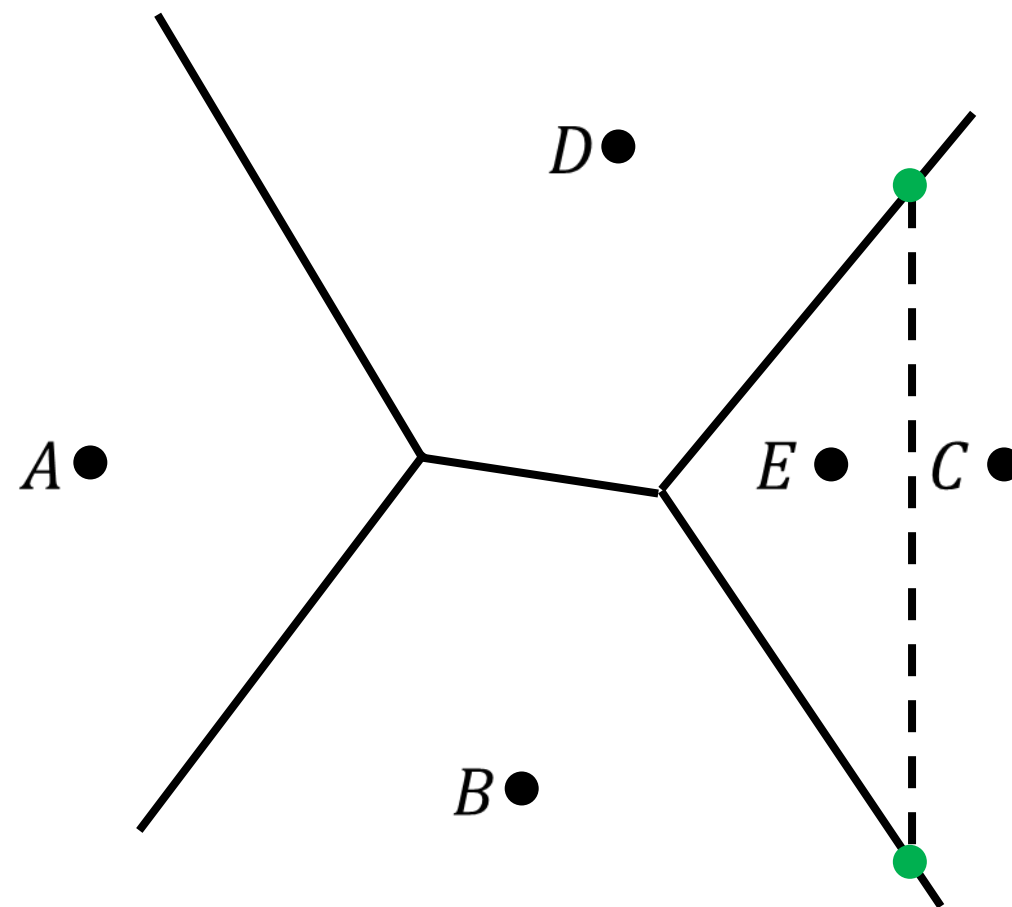
Adding a New Site

- Since E is inside cell C , construct the segment of the perpendicular bisector $PB(C, E)$ that lies inside cell C



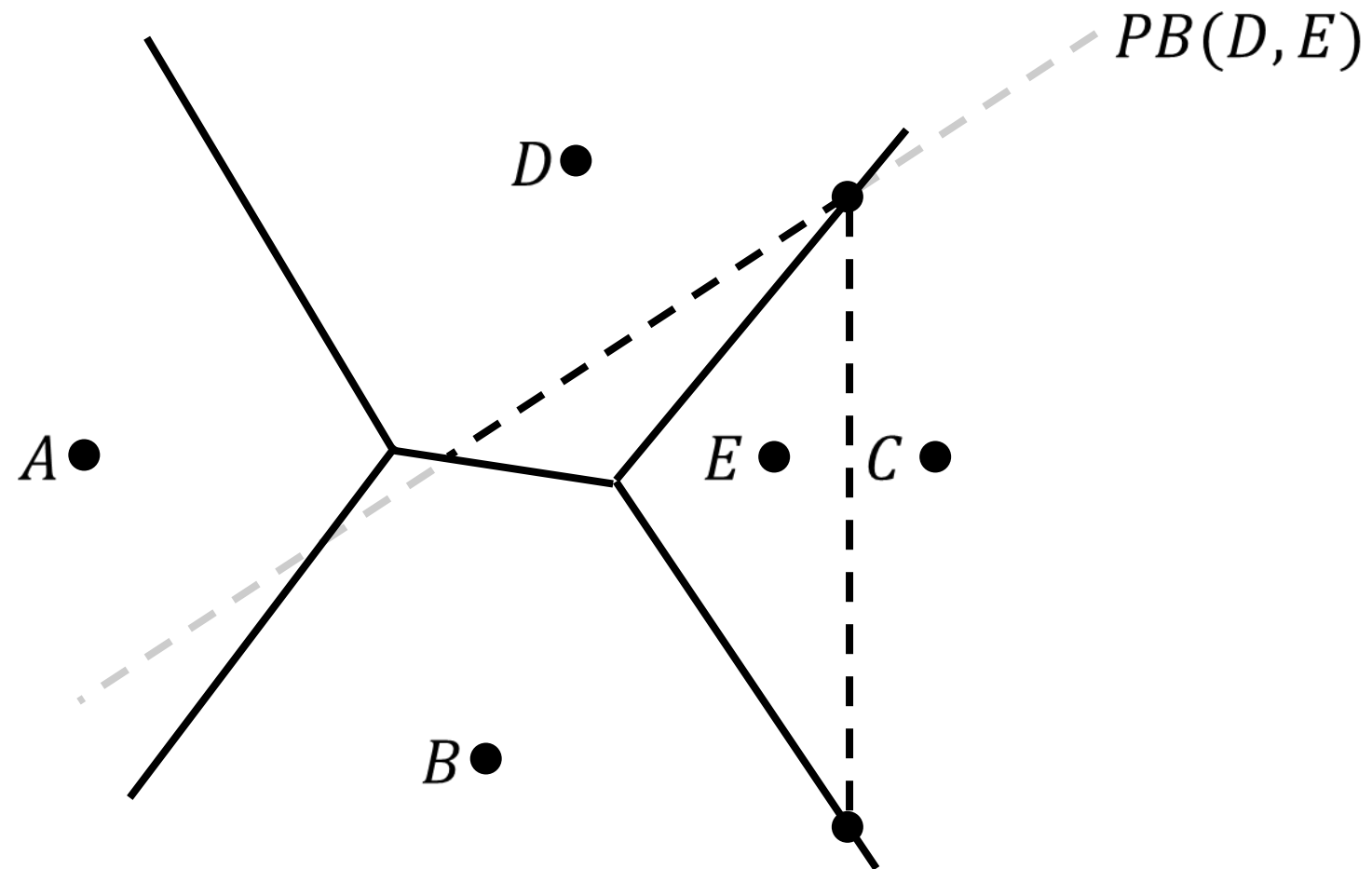
Adding a New Site

- Notice that we have 2 new **vertices**. Select a vertex to process first, so in this case, let's choose the vertex adjacent to site D



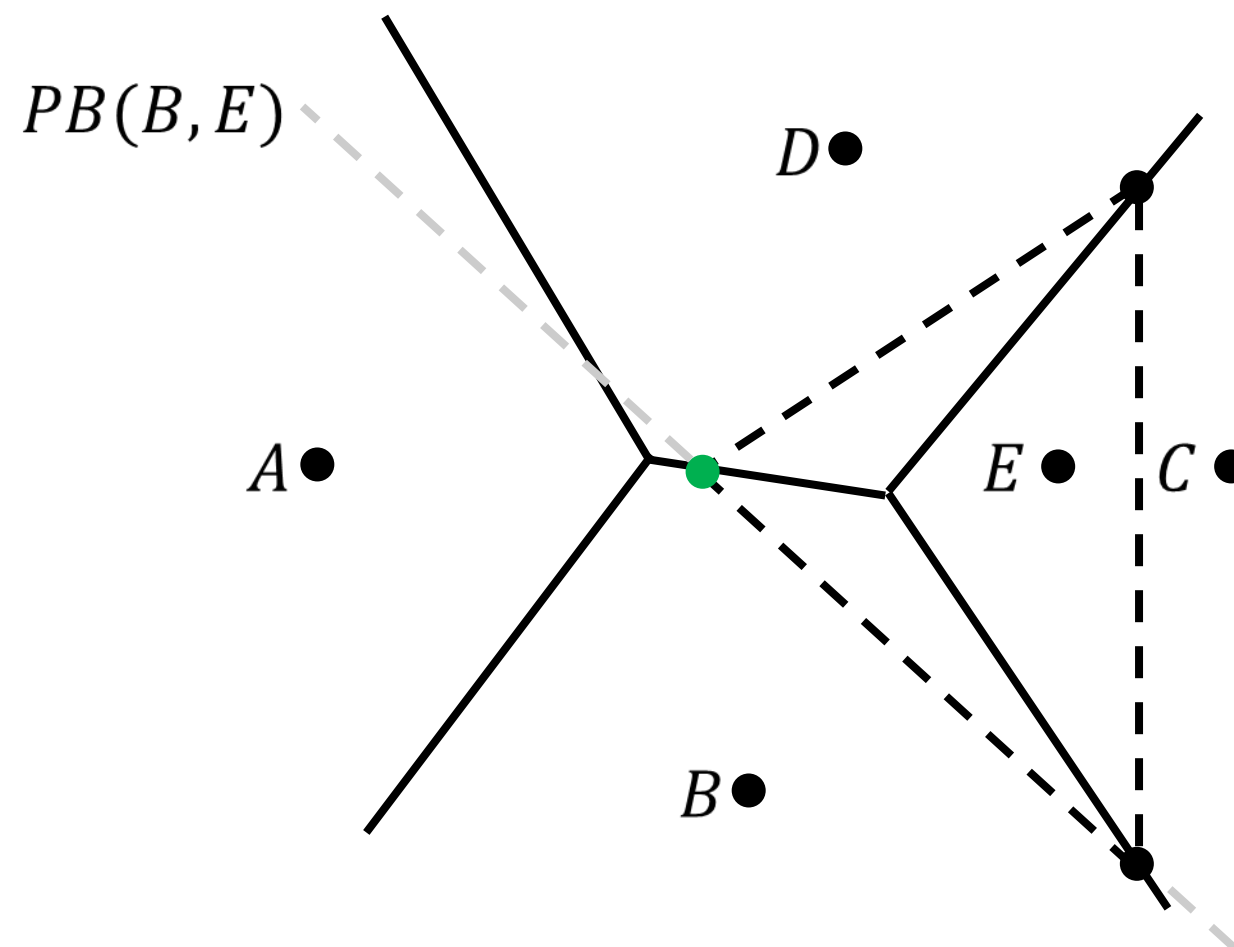
Adding a New Site

- Construct $PB(D, E)$



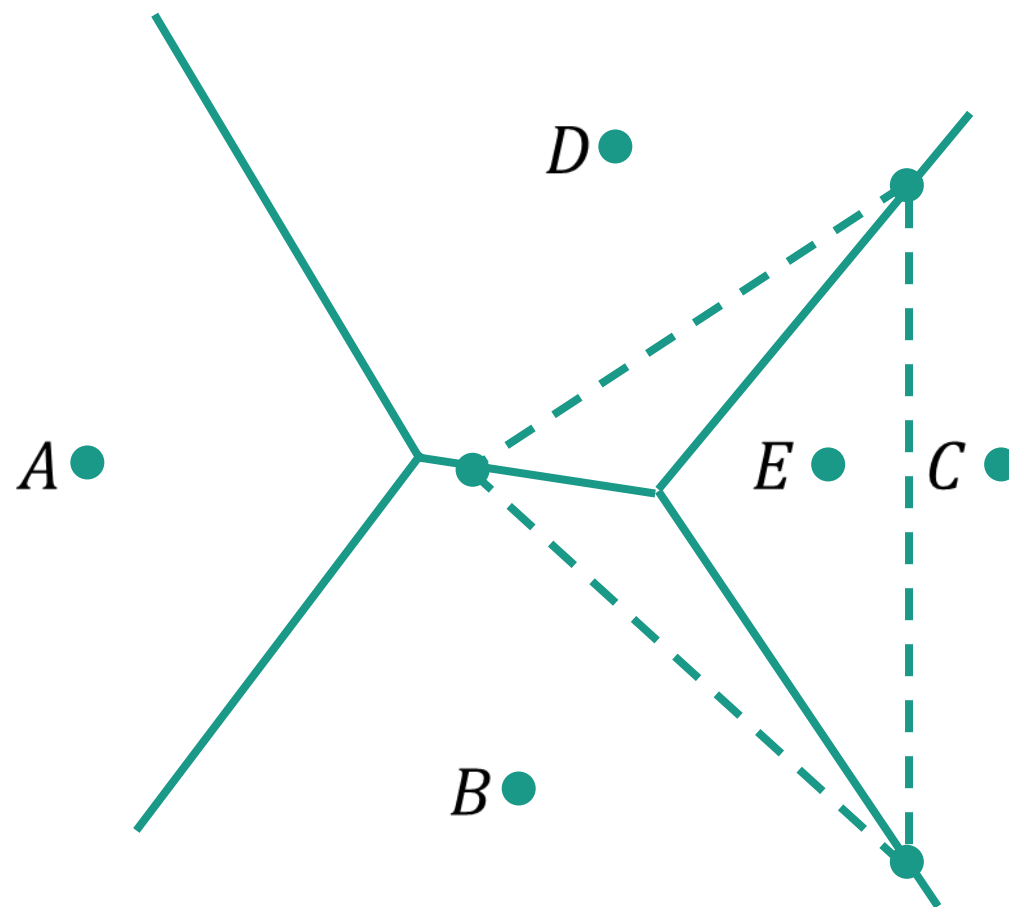
Adding a New Site

- Notice that we created a new **vertex** adjacent to B , so we construct $PB(B, E)$



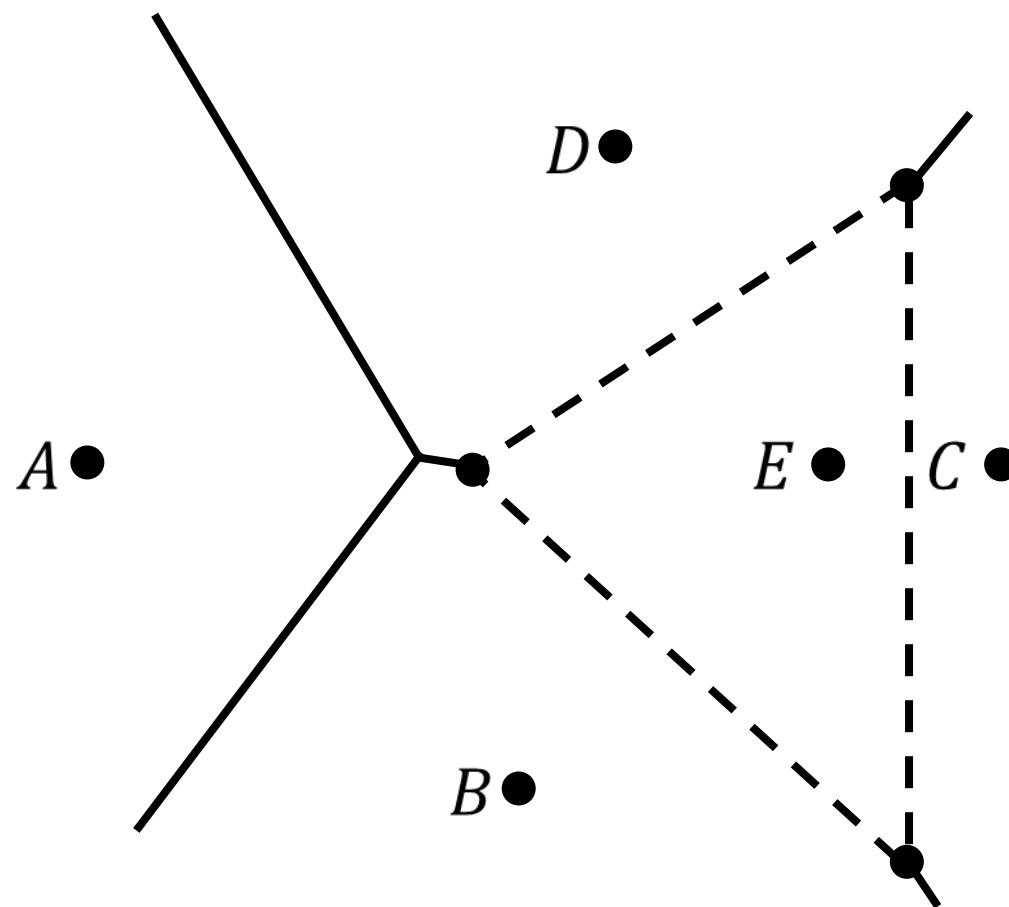
Adding a New Site

- Notice that the segment passed through a vertex that we created before, so we did not really create a new vertex, so we stop here



Adding a New Site

- Finally, remove the segments of existing edges that are inside the new cell for site E

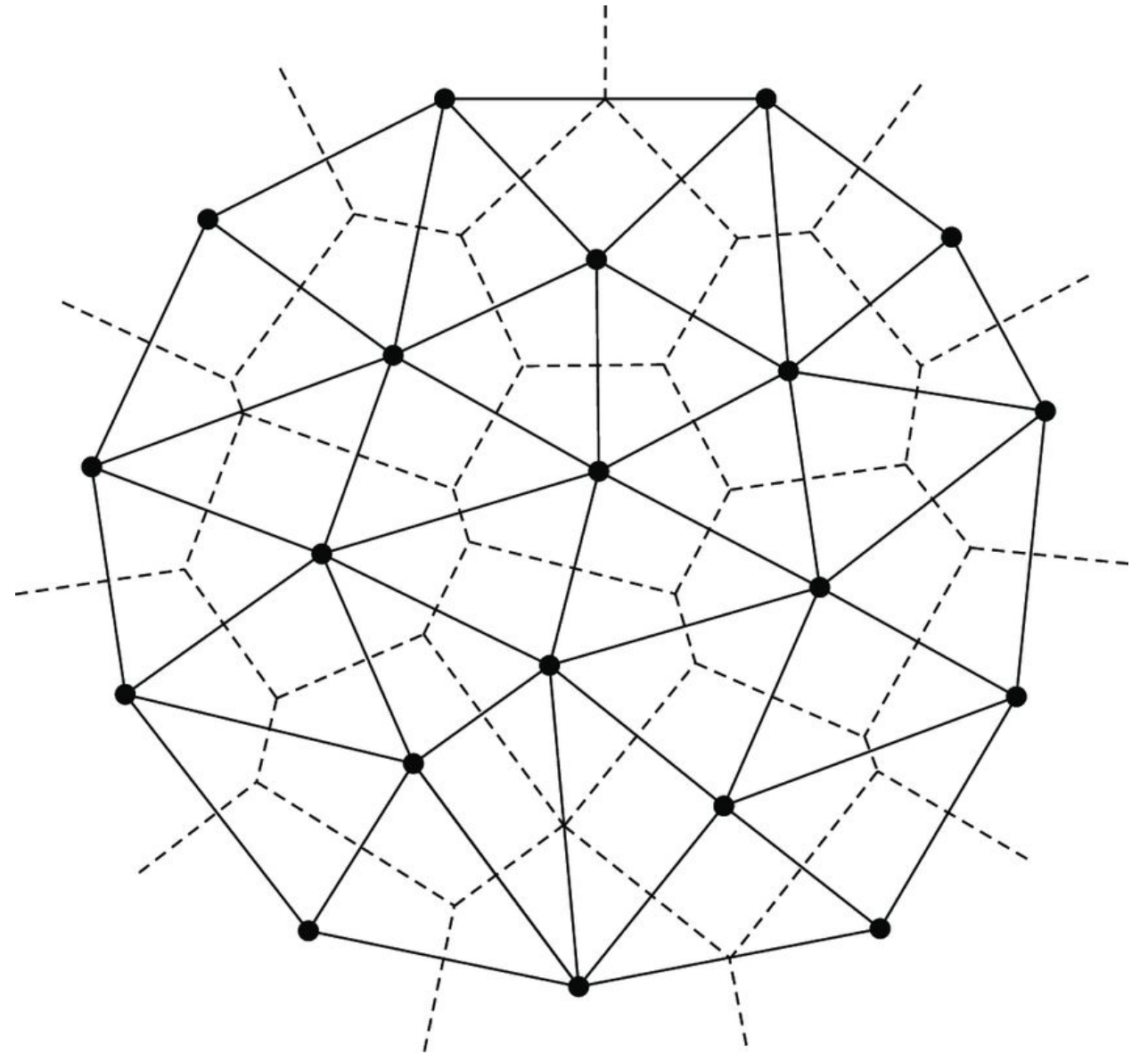


Constructing Voronoi Diagrams

- ▶ Voronoi diagrams can be constructed by computing the intersections of the half-planes generated from the perpendicular bisectors in $O(n^2 \log n)$ time
- ▶ Fortune's algorithm can construct a Voronoi diagram in $O(n \log n)$ time, which involves a horizontal sweeping line going from top to bottom, constructing parabolic arcs at certain event points
 - ▶ Intersections of parabolic arcs define the edges
 - ▶ <https://youtu.be/k2P9yWSMaXE>

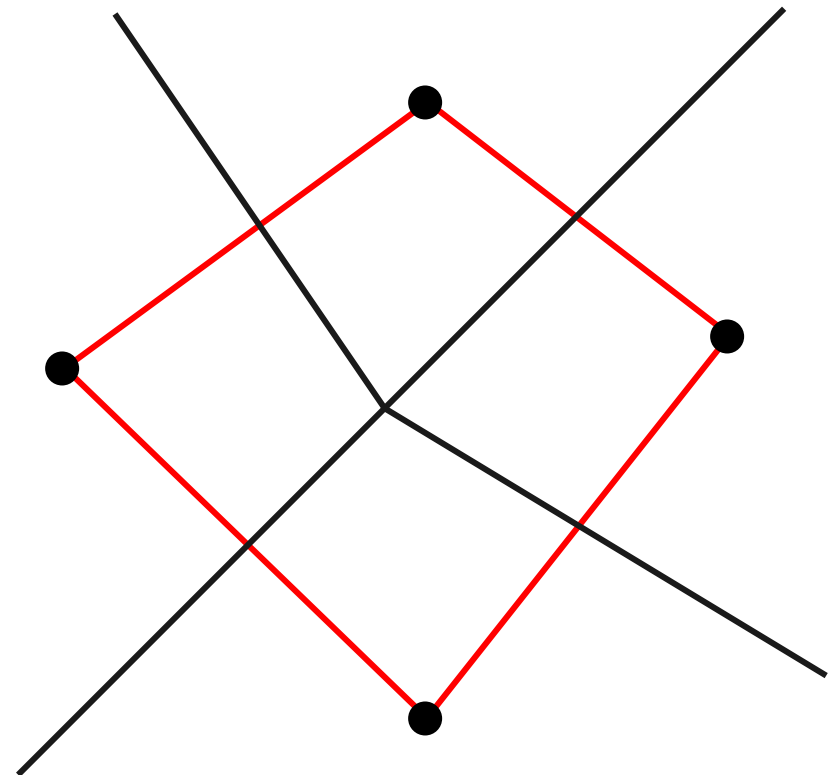
Delaunay Graph

- Given a Voronoi diagram, if you connect adjacent sites with edges, we end up with a graph called the *Delaunay graph*, named after Russian mathematician Boris Nikolaevich Delone



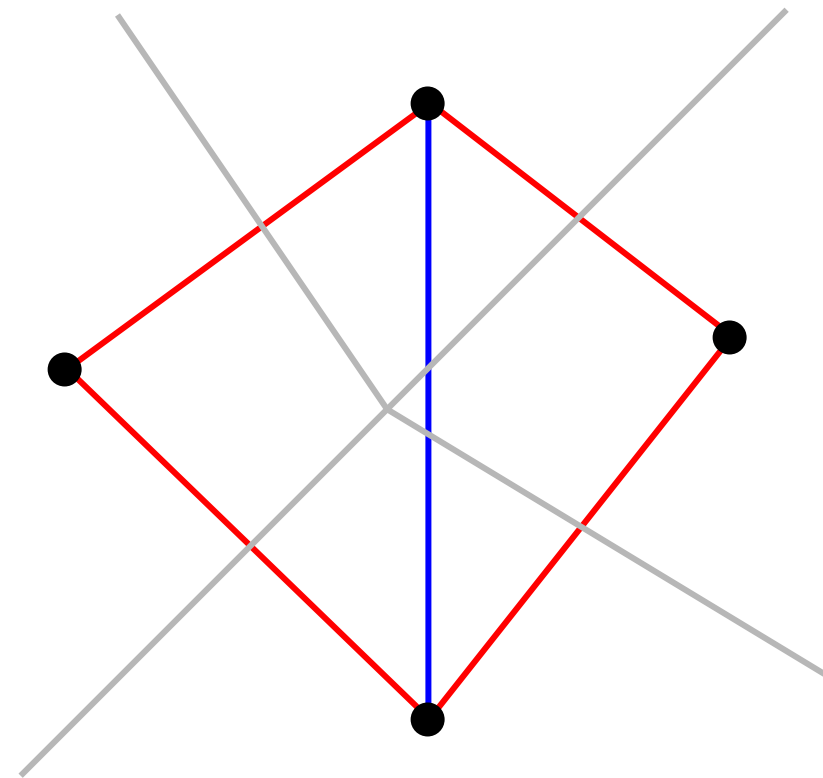
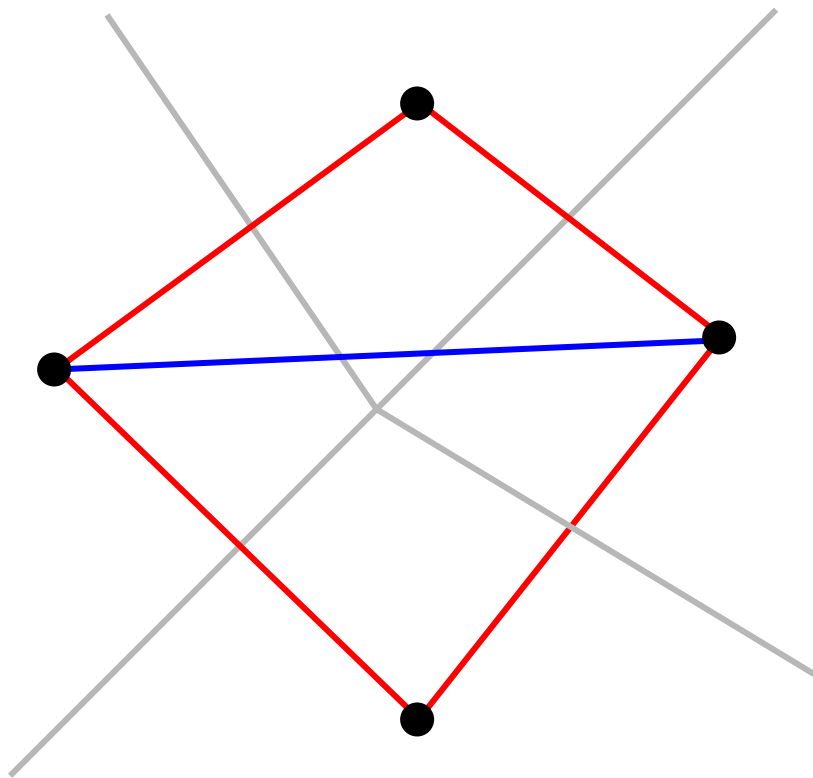
Delaunay Graph

- ▶ Although the Delaunay graph looks like a triangulation, there are rare cases where you have a Voronoi vertex with degree four, which means you have a quad rather than a triangle



Delaunay Graph

- We can add an edge to split this quad into two triangles, but how do we choose which edge to use? Which is "better"?



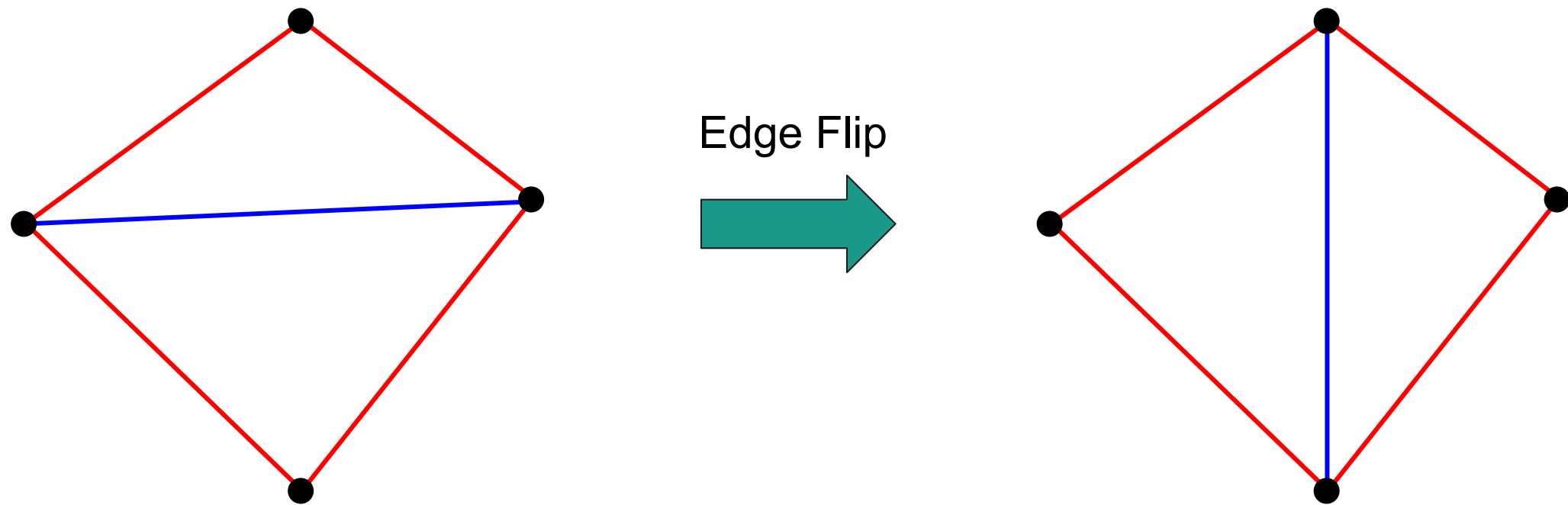
Angle-Optimal Triangulation

- ▶ Given a triangulation T of a set of points P that consists of m triangles, the *angle-vector* of T , denoted by $A(T) = (a_1, a_1, \dots, a_{3m})$ is the sequence of all $3m$ angles (3 angles per triangle) in the triangulation, sorted in increasing order.
- ▶ Given two triangulations T and T' , as well as their angle-vectors $A(T)$ and $A(T')$, $A(T) > A(T')$ if $A(T)$ is lexicographically larger than $A(T')$, i.e., there exists an index $1 \leq i \leq 3m$ such that:

$$a_j = a'_j \text{ for all } j < i, \text{ and } a_i > a'_i$$

Angle-Optimal Triangulation

- ▶ A triangulation T is angle-optimal if $A(T) \geq A(T')$ for all triangulations T' of P
- ▶ Going back to the original problem, we can start with any triangulation T and get $A(T)$, and do an *edge flip* to get T' and $A(T')$, and check if $A(T) > A(T')$

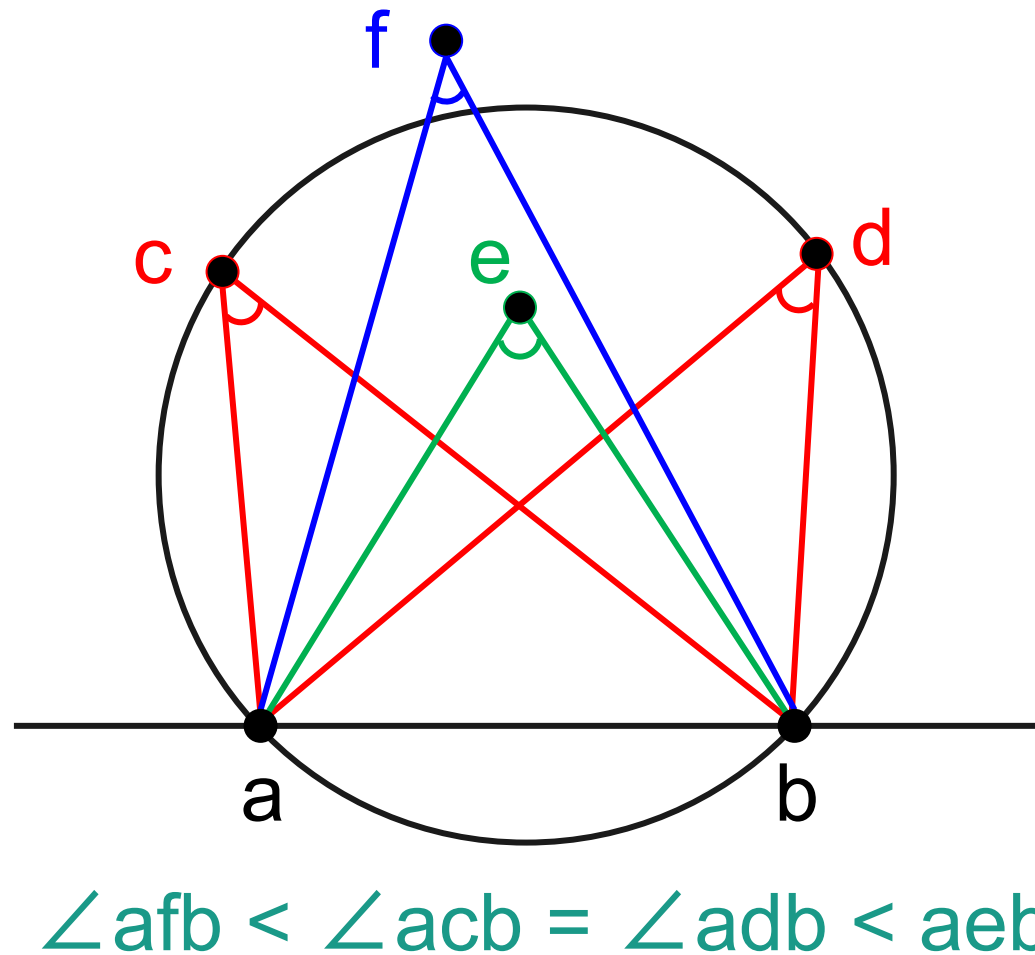


Angle-Optimal Triangulation

- ▶ Let e be an edge in a triangulation T , and let T' be the resulting triangulation after an edge flip. The edge e is *illegal* if $A(T) < A(T')$
- ▶ A triangulation T is *legal* if there are no illegal edges in T , and it follows that any angle-optimal triangulation is a legal triangulation
- ▶ Given an initial triangulation T , we can turn T into a legal triangulation by flipping edges until all edges are legal

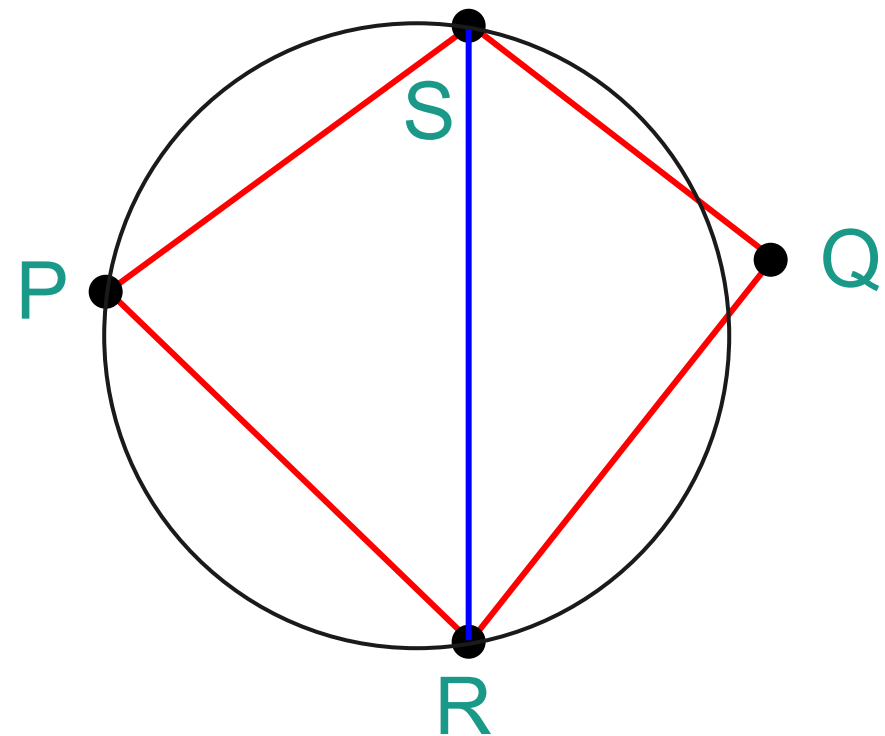
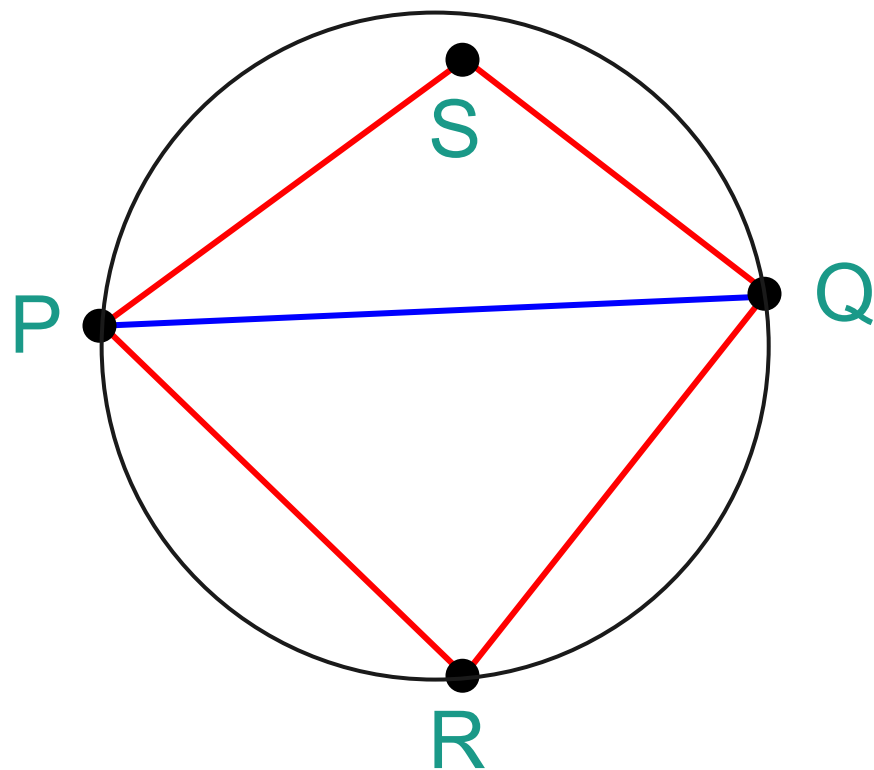
Angle-Optimal Triangulation

- It turns out that we do not need to check all the angles in $A(T)$ and $A(T')$ to check if $A(T) > A(T')$
- We can make use of Thales' Theorem



Angle-Optimal Triangulation

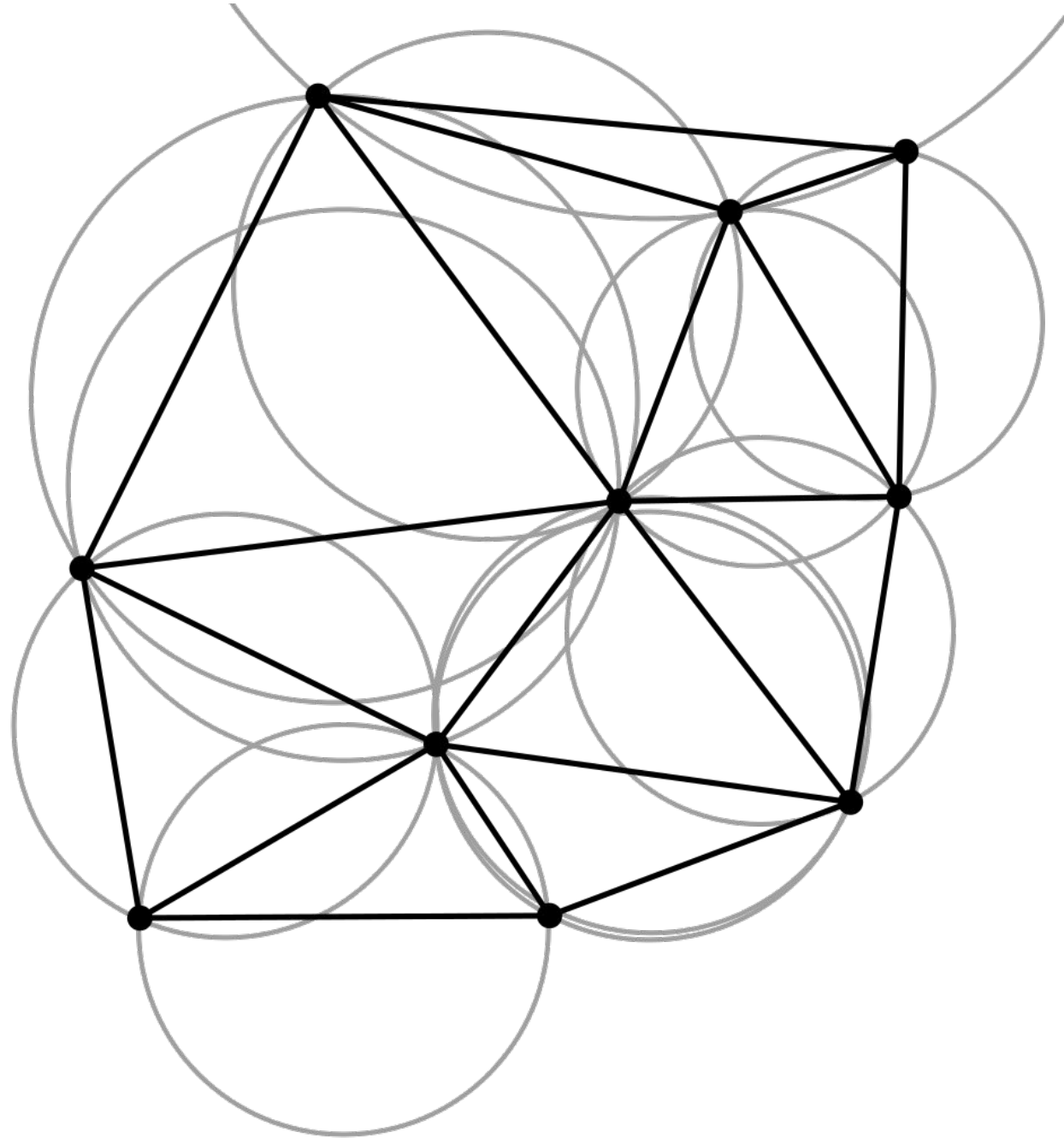
- Given an edge PQ that is incident to triangles PQR and PQS , and a circle C that is the circle defined by P , Q , and R (i.e., the circumcircle of PQR), the edge PQ is illegal if and only if S lies in the interior of the circle C



Delaunay Triangulation

- ▶ Going back to our Delaunay Graph, we can transform a Delaunay Graph into a Delaunay Triangulation by adding edges to the Delaunay Graph to make all faces to be triangles
- ▶ A triangulation T of a set of points P is a Delaunay triangulation if and only if the circumcircle of any triangle in T does not contain a point of P
- ▶ Sound familiar?
- ▶ We need to add legal edges to transform a Delaunay Graph into a Delaunay Triangulation

Delaunay Triangulation



https://en.wikipedia.org/wiki/Delaunay_triangulation#/media/File:Delaunay_circumcircles_vectorial.svg

Delaunay Triangulation

- ▶ A Delaunay Triangulation is a legal triangulation, and any legal triangulation is a Delaunay Triangulation
- ▶ If all vertices of the resulting Voronoi Diagram of a point set P has a maximum degree of three, then P is said to be in a general position, and the dual graph is a Delaunay Triangulation
- ▶ The resulting triangulation is also unique

Delaunay Triangulation

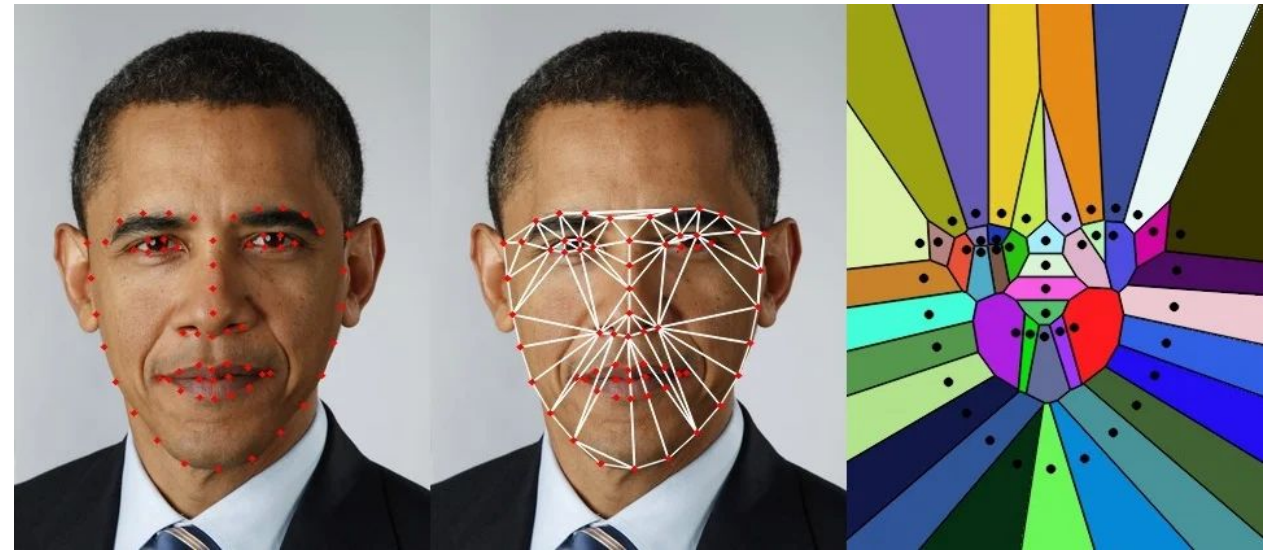
- ▶ Given a set of points P , a Delaunay Triangulation of P can be constructed using the Bowyer–Watson algorithm, an iterative algorithm where you construct a legal triangulation one point at a time
- ▶ A demonstration can be found here:
<https://www.youtube.com/watch?v=GctAunEuHt4>

Duality

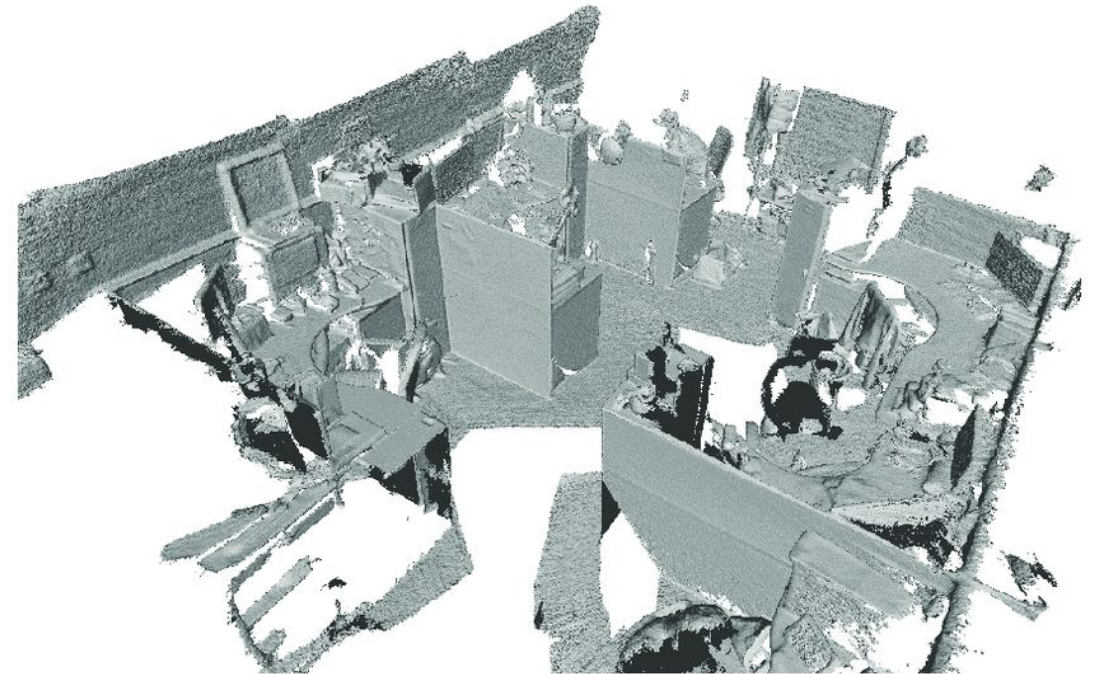
- ▶ Given a Voronoi Diagram, a Delaunay Graph can be obtained by creating an edge between adjacent sites
- ▶ Given a Delaunay Graph/Triangulation, you can obtain the sites in a Voronoi Diagram via the vertices in the Delaunay Graph, and the vertices in a Voronoi Diagram via the circumcenters of the Delaunay triangles
- ▶ As such, Delaunay Graphs and Voronoi Diagrams are *duals* of each other

Voronoi Diagrams and Delaunay Triangulation Applications

- ▶ Spatial queries
- ▶ Image compression
- ▶ Face morphing
- ▶ Procedural Map Generation
- ▶ 3D surface reconstruction from a point cloud
- ▶ and others



<https://learnopencv.com/delaunay-triangulation-and-voronoi-diagram-using-opencv-c-python/>



https://www.researchgate.net/figure/3D-reconstruction-result-by-InfiniTAM_fig4_326137545